Model : 223II0

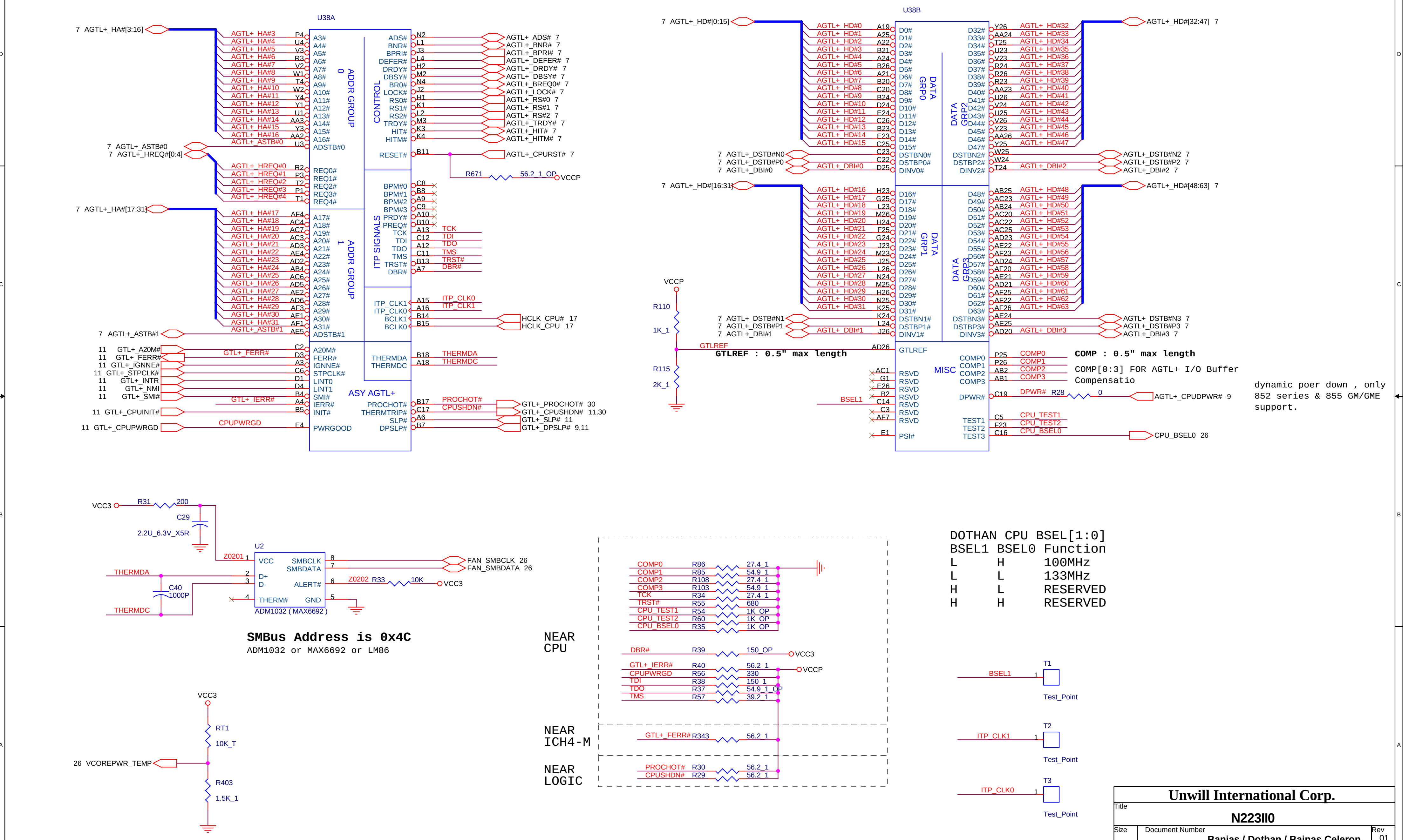
Intel Banias / Dothan with Intel 855GME Chipset

Revision History

A	12/2003	ORIGINAL RELEASE
B	03/2004	ORIGINAL RELEASE
c	04/2004	ORIGINAL RELEASE
1	05/2004	ORIGINAL RELEASE

Unwill International Corp.

Title	223II0	
Size	Document Number 2665 INDEX	Rev 01
Date: Tuesday, May 25, 2004	Sheet 1 of 39	

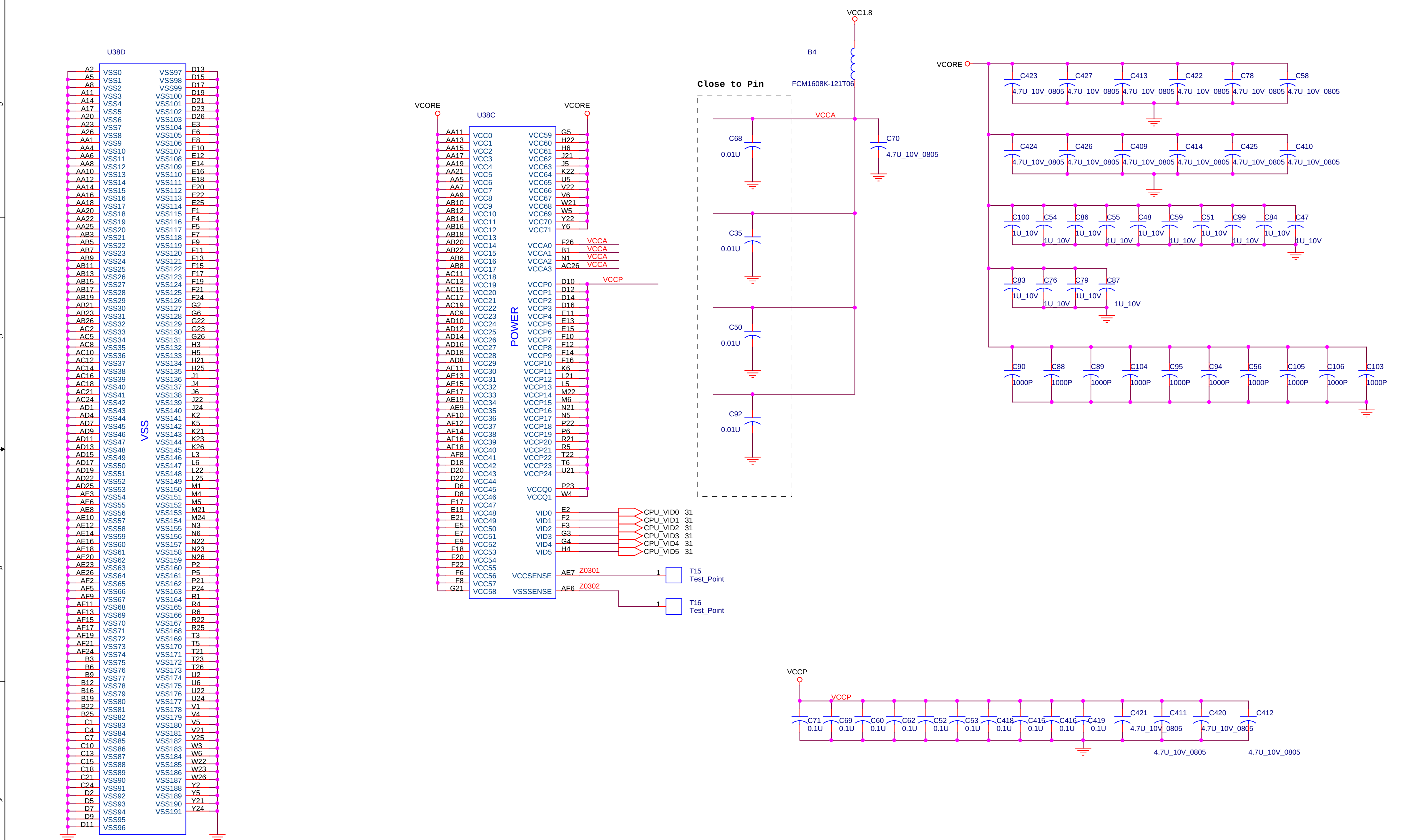


Unwill International Corp.

Title: **N223II0**

Size	Document Number	Rev
	2665 **Banias / Dothan / Bainas Celeron**	01

Date: Tuesday, May 25, 2004 | Sheet 2 of 39

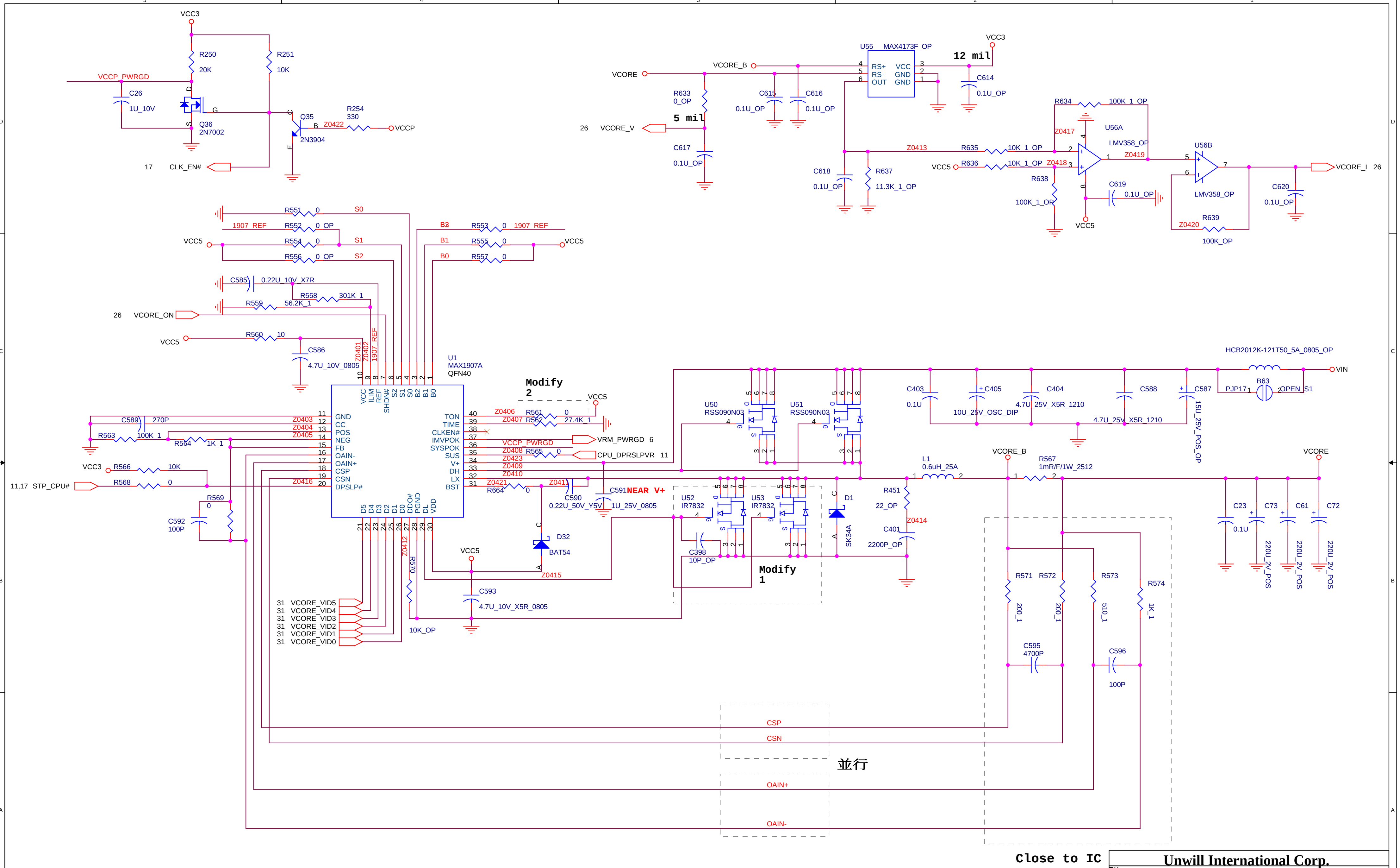


Unwill International Corp.

Title: N223II0

Size	Document Number	Rev
	2665 CPU Power	01

Date: Tuesday, May 25, 2004 Sheet 3 of 39



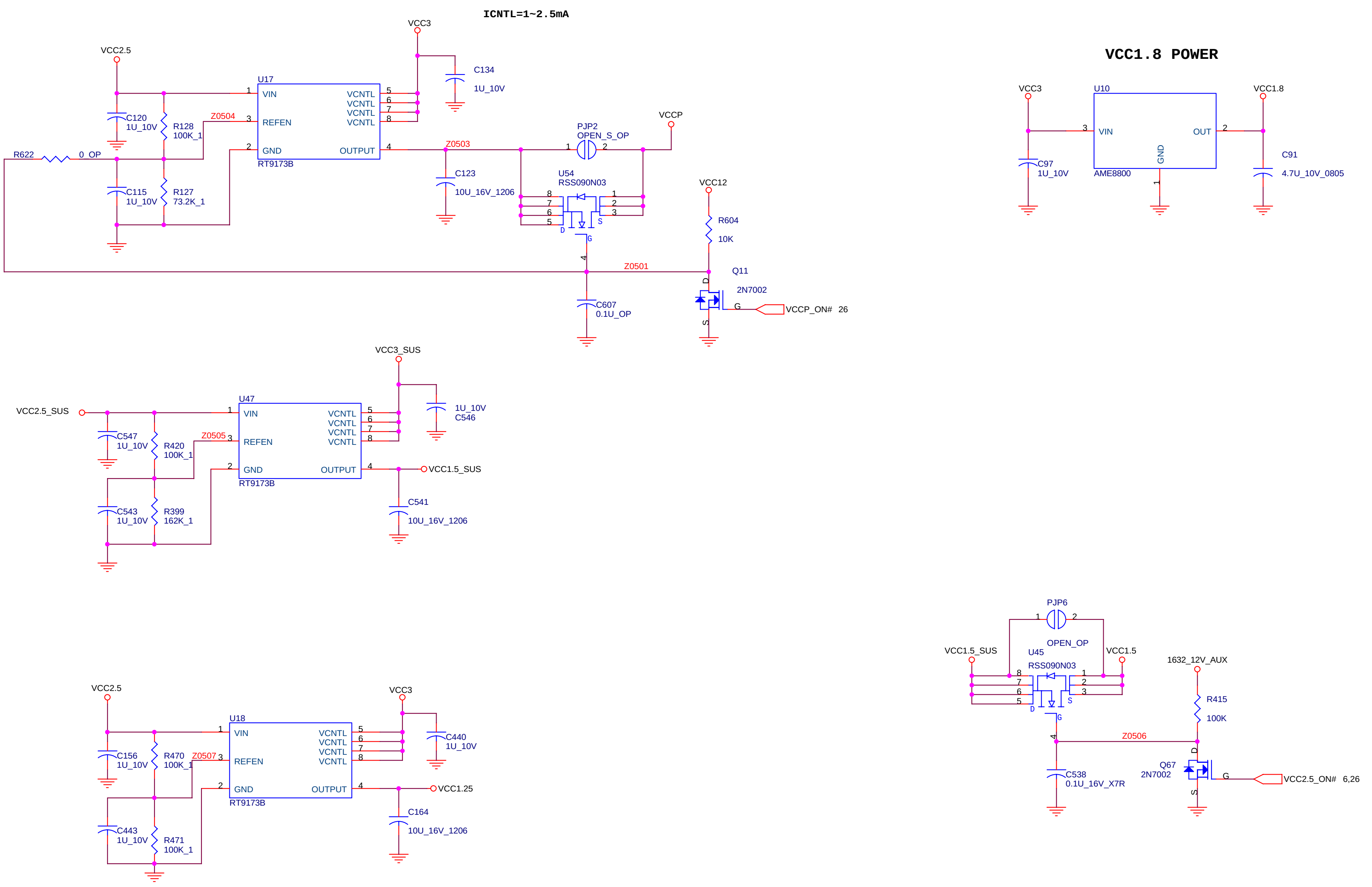
Unwill International Corp.

Title: N223II0

Size | Document Number: CPU Power MAX1907 | Rev 01

2665

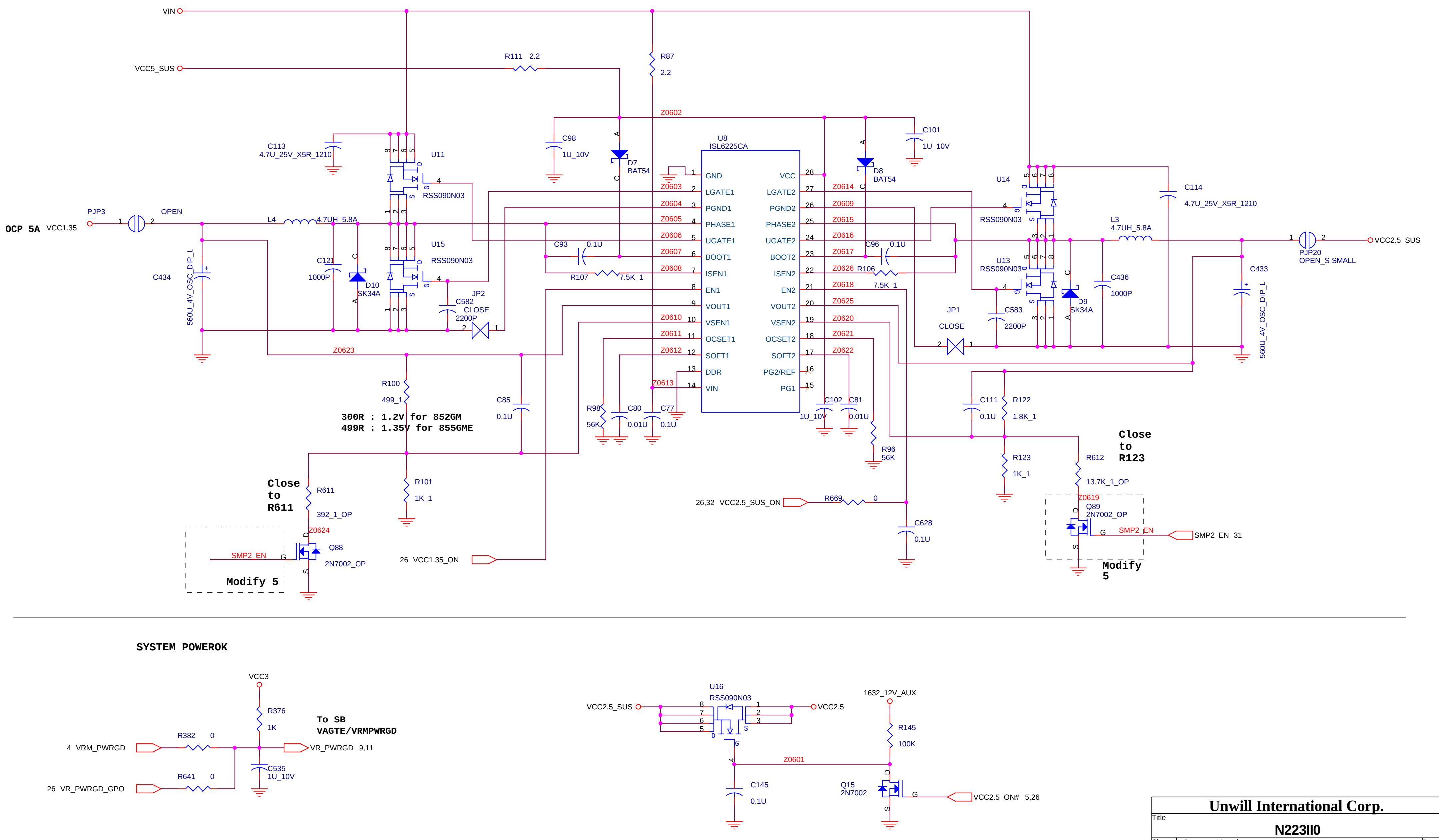
Date: Tuesday, May 25, 2004 | Sheet 4 of 39



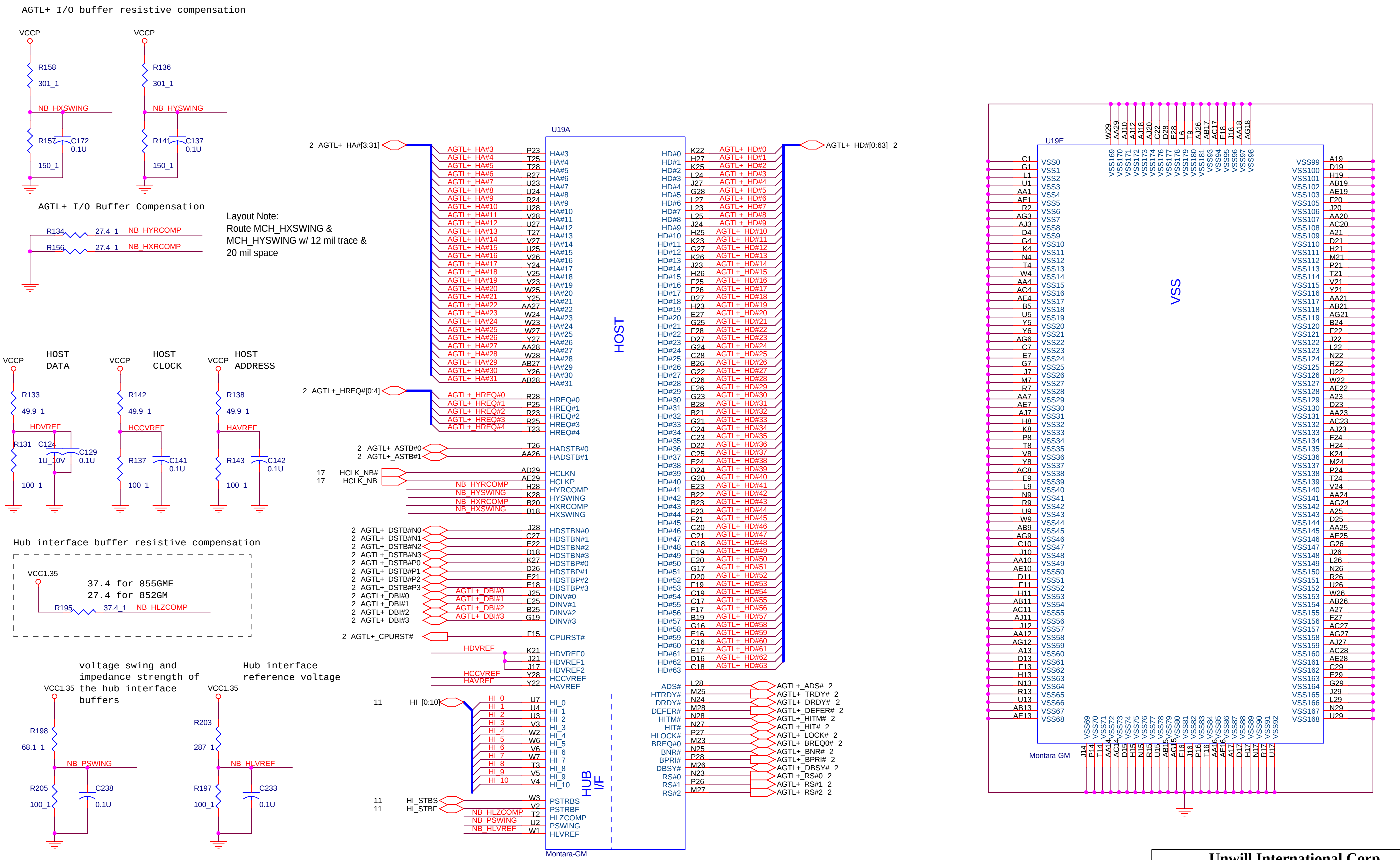
Unwill International Corp.

Title	N223II0
Size	Document Number: 2665 — 1.05V & 1.2V/ 1.35V — Rev 01
Date: Tuesday, May 25, 2004	Sheet 5 of 39

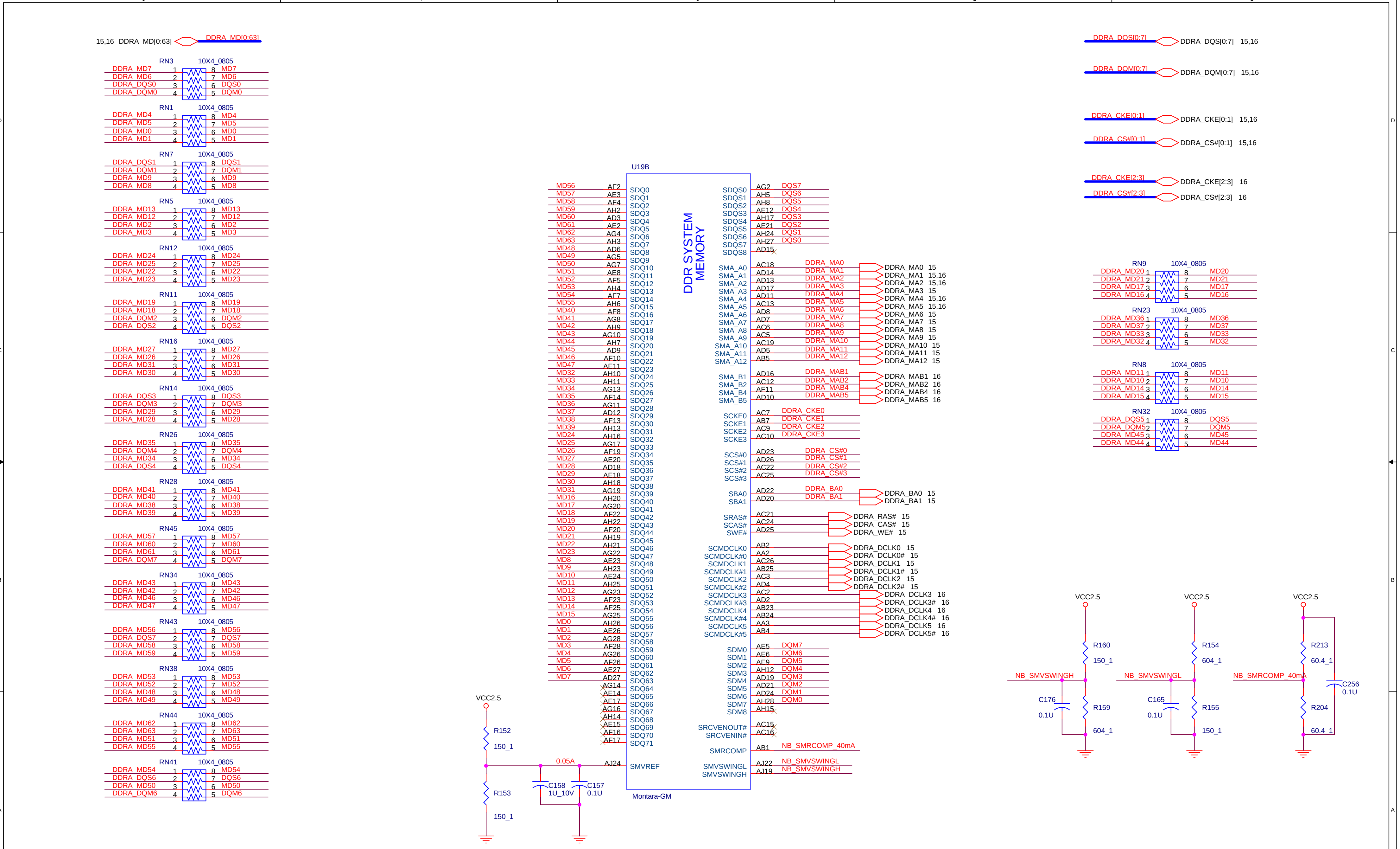
VCC 2.5/(1.35V OR 1.2V) POWER



Unwill International Corp.	
Title	N223II0
Size / Document Number: 2665	1.5V / 2.5V / PWROK — Rev 01
Date: Tuesday, May 25, 2004	Sheet 6 of 39

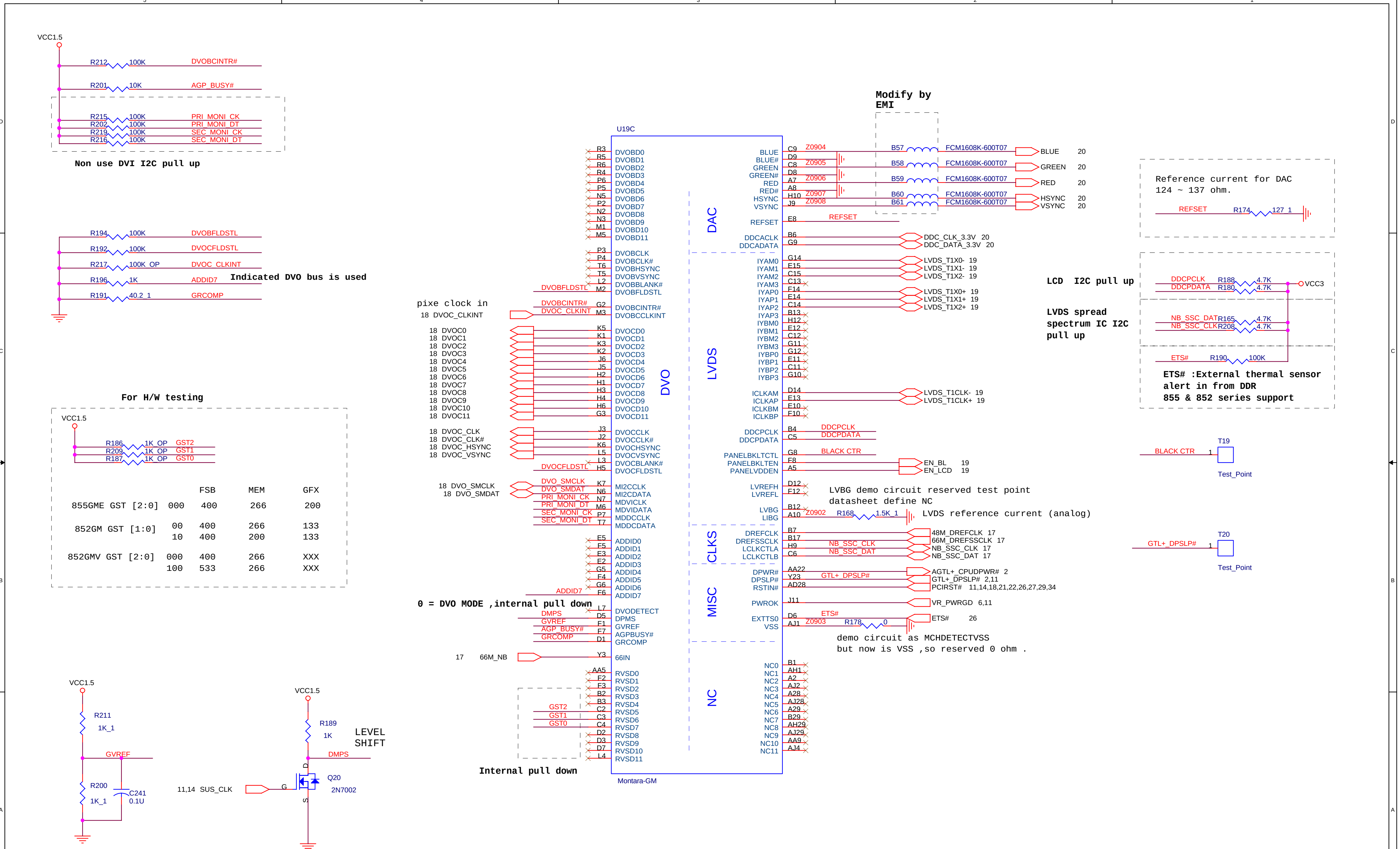


Unwill International Corp.	
Title	N223II0
Size / Document Number	NB HOST — 2665
Rev	01
Date:	Tuesday, May 25, 2004
Sheet	7 of 39



Unwill International Corp.

Title	N223II0	
Size	Document Number 2665 NB MEM	Rev 01
Date: Tuesday, May 25, 2004	Sheet 8 of 39	



Unwill International Corp.

Title	N223II0	
Size	Document Number: 2665 NB DAC	Rev 01
Date: Tuesday, May 25, 2004	Sheet 9 of 39	

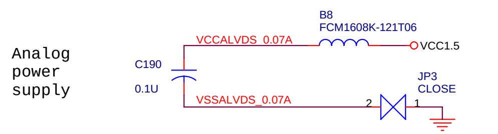
VCC1.35 :

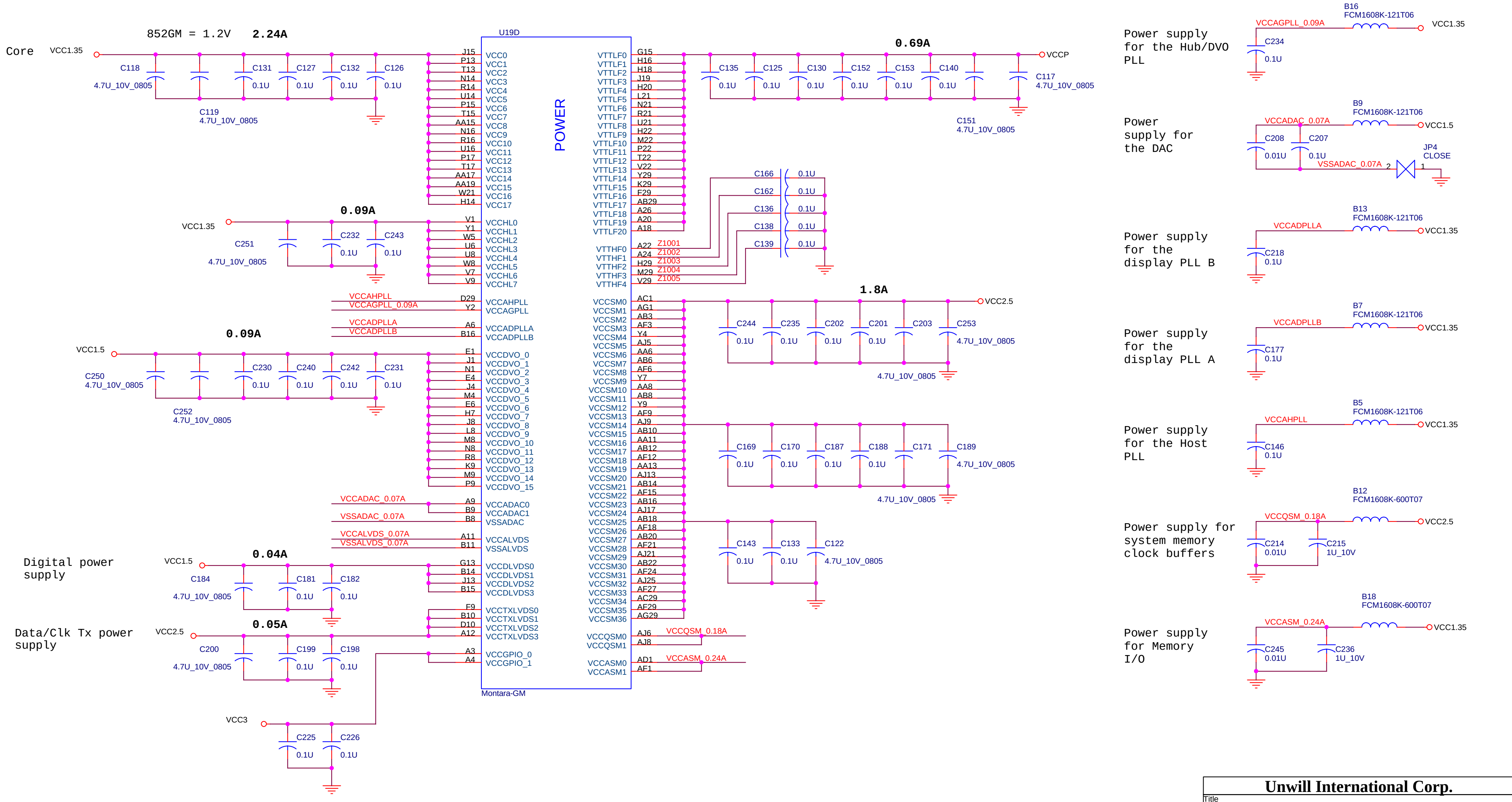
	VCC1.35	
855GME	1.35V	
852GMV	1.5V	
852GM	1.2V	

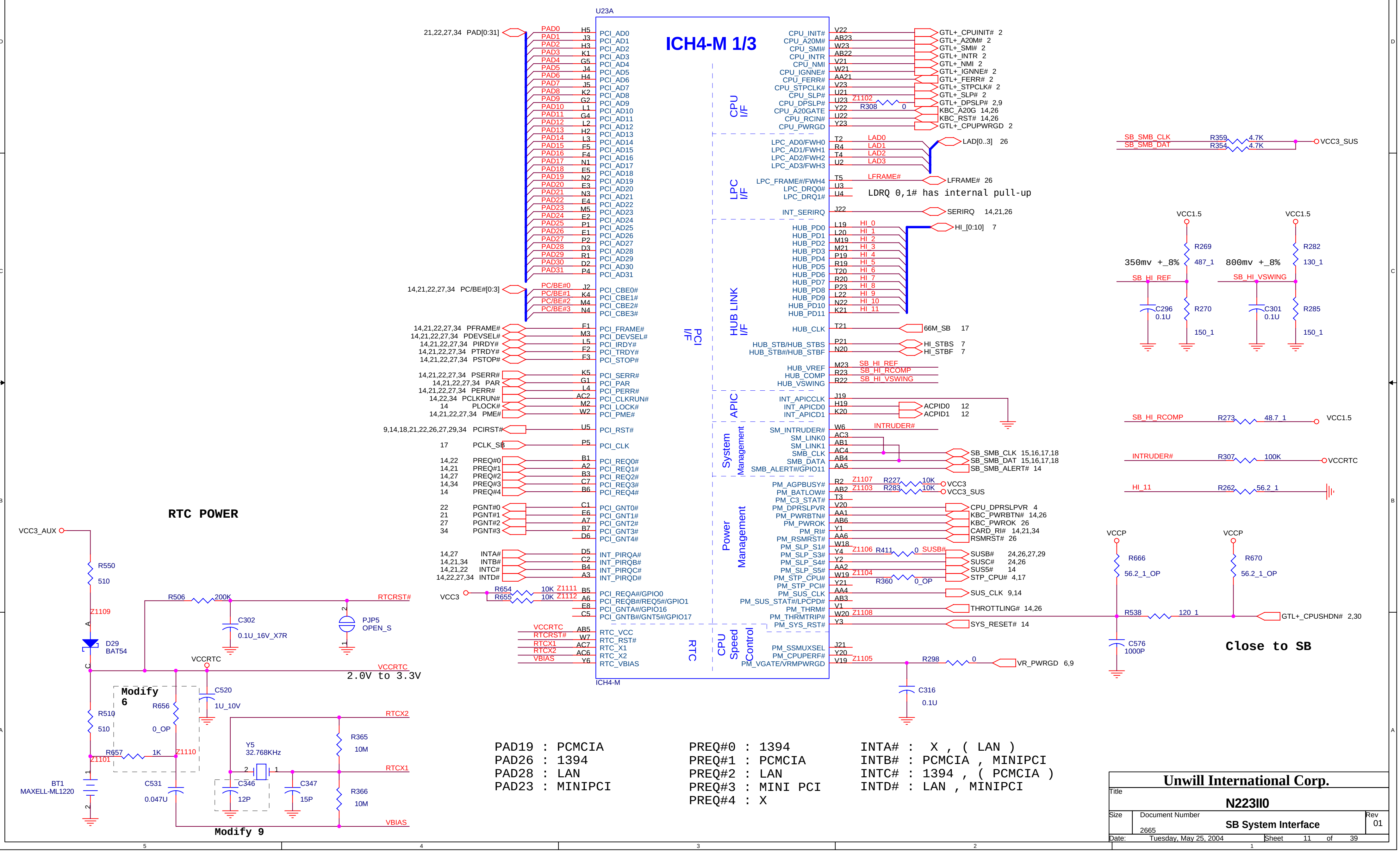
VTTLF is the low frequency connection from the board. This signal is the primary connection of power for GMCH.

VTTHF is the high frequency supply. It is for direct connection from an internal package plane to a capacitor placed immediately adjacent to the GMCH.
NOTE: Not to be connected to power rail.

Analog Power





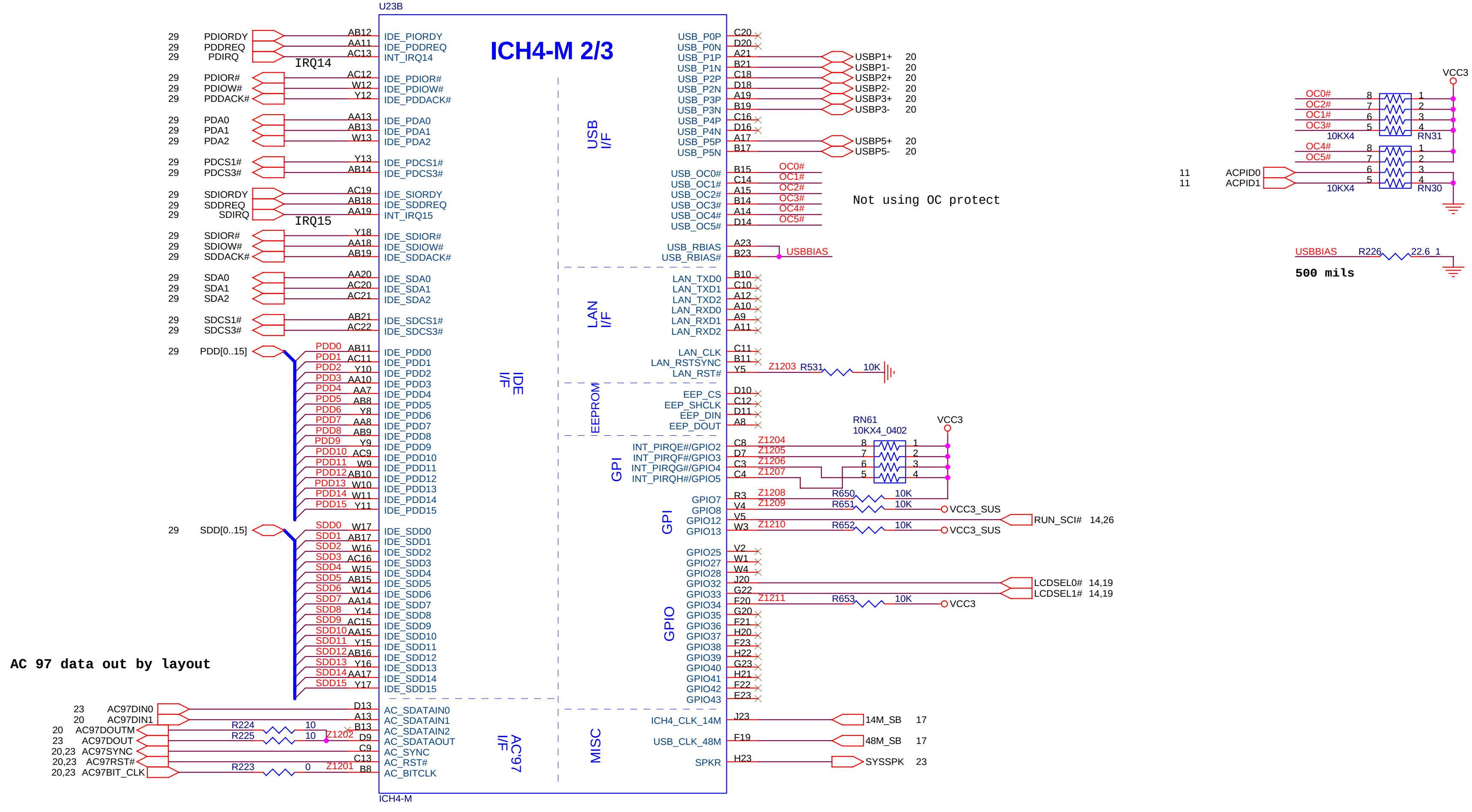


PAD19 : PCMCIA
PAD26 : 1394
PAD28 : LAN
PAD23 : MINIPCI

PREQ#0 : 1394
PREQ#1 : PCMCIA
PREQ#2 : LAN
PREQ#3 : MINI PCI
PREQ#4 : X

INTA# : X , (LAN)
INTB# : PCMCIA , MINIPCI
INTC# : 1394 , (PCMCIA)
INTD# : LAN , MINIPCI

Unwill International Corp.	
Title	N223II0
Size / Document Number	2665 — SB System Interface
Rev	01
Date:	Tuesday, May 25, 2004
Sheet	11 of 39

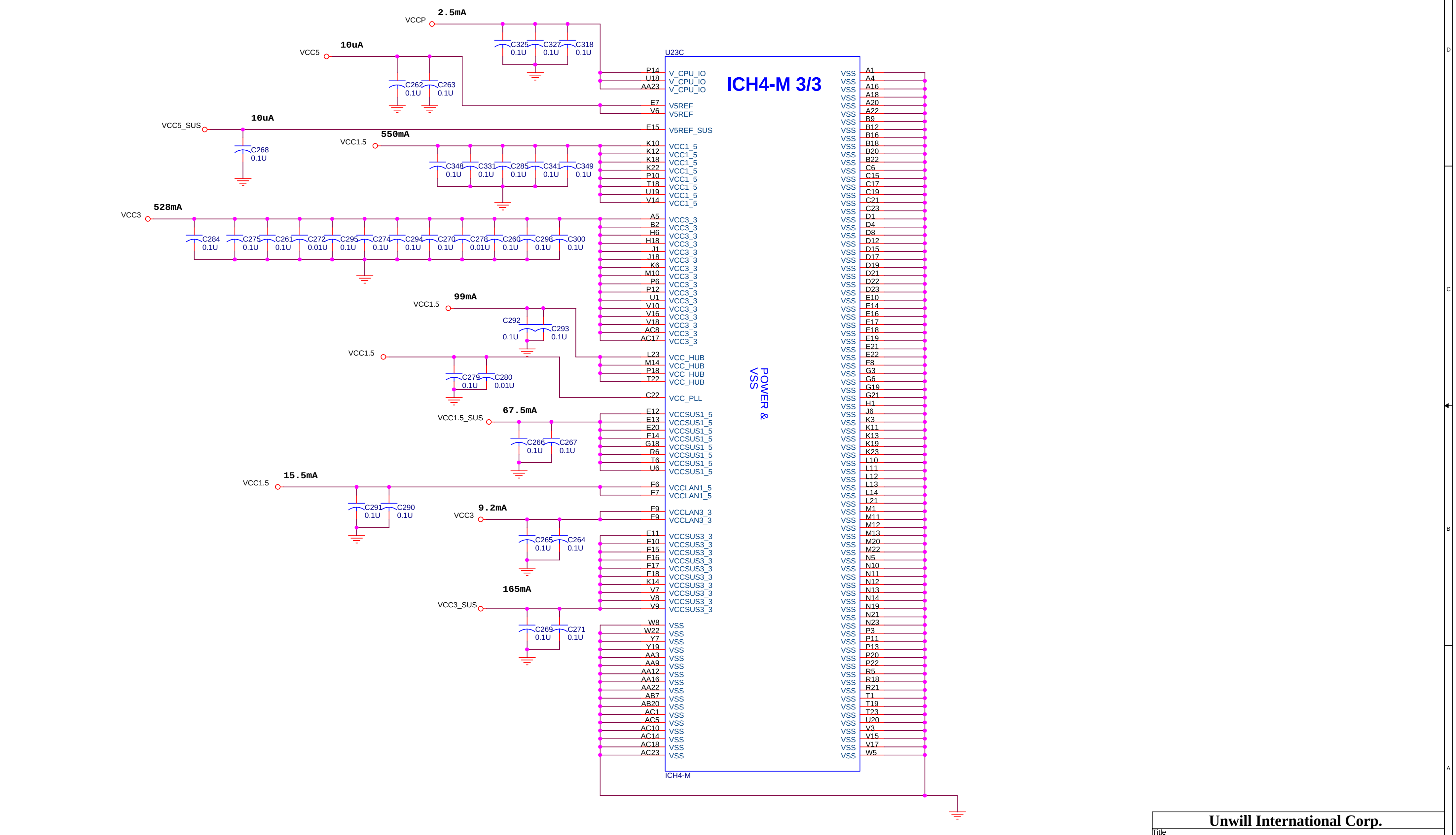


Unwill International Corp.

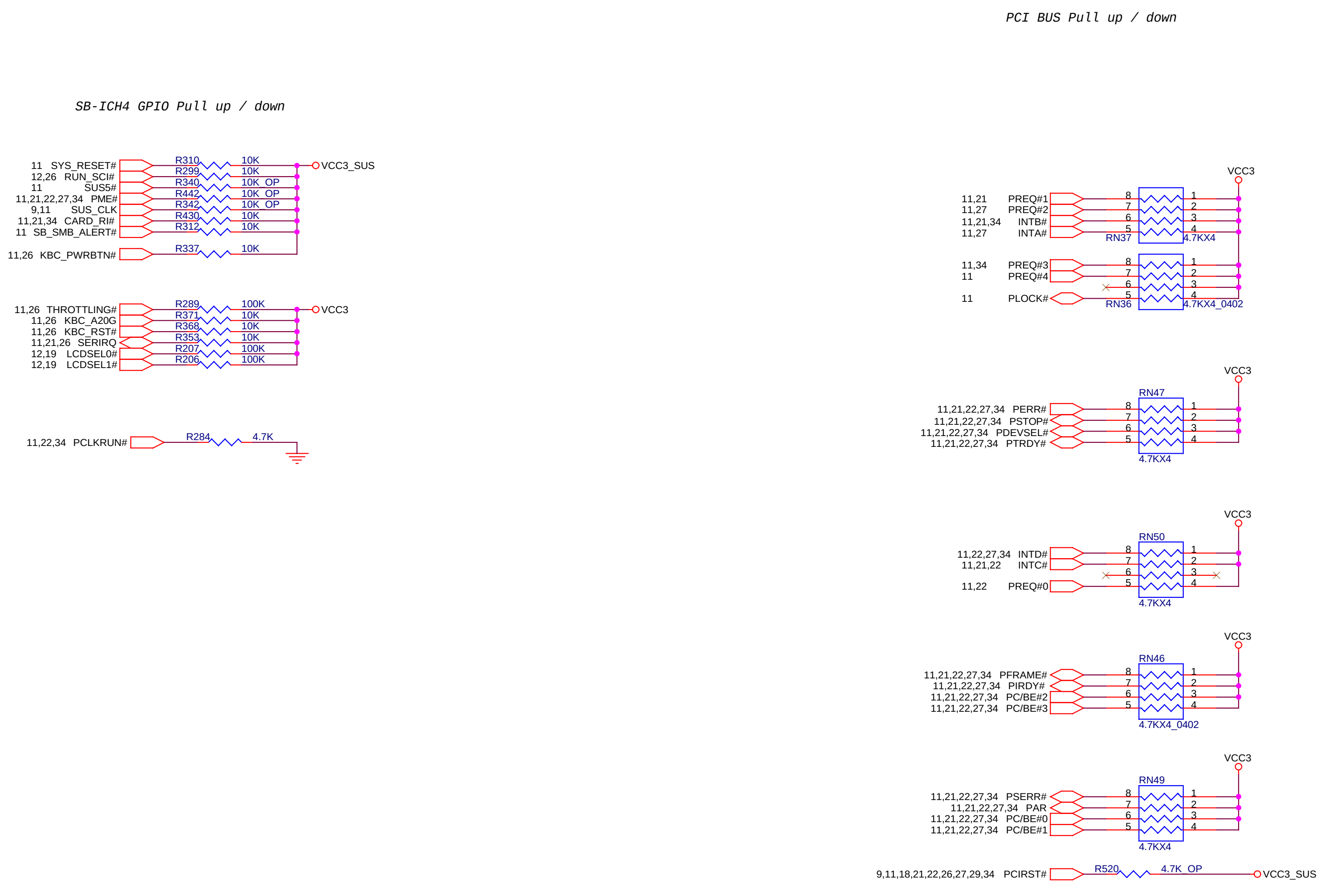
Title: N223II0

Size | Document Number: 2665 | SB I/O Interface | Rev 01

Date: Tuesday, May 25, 2004 | Sheet 12 of 39



Unwill International Corp.			
Title	N223II0		
Size	Document Number 2665	SB Power	Rev 01
Date: Tuesday, May 25, 2004		Sheet 13 of 39	

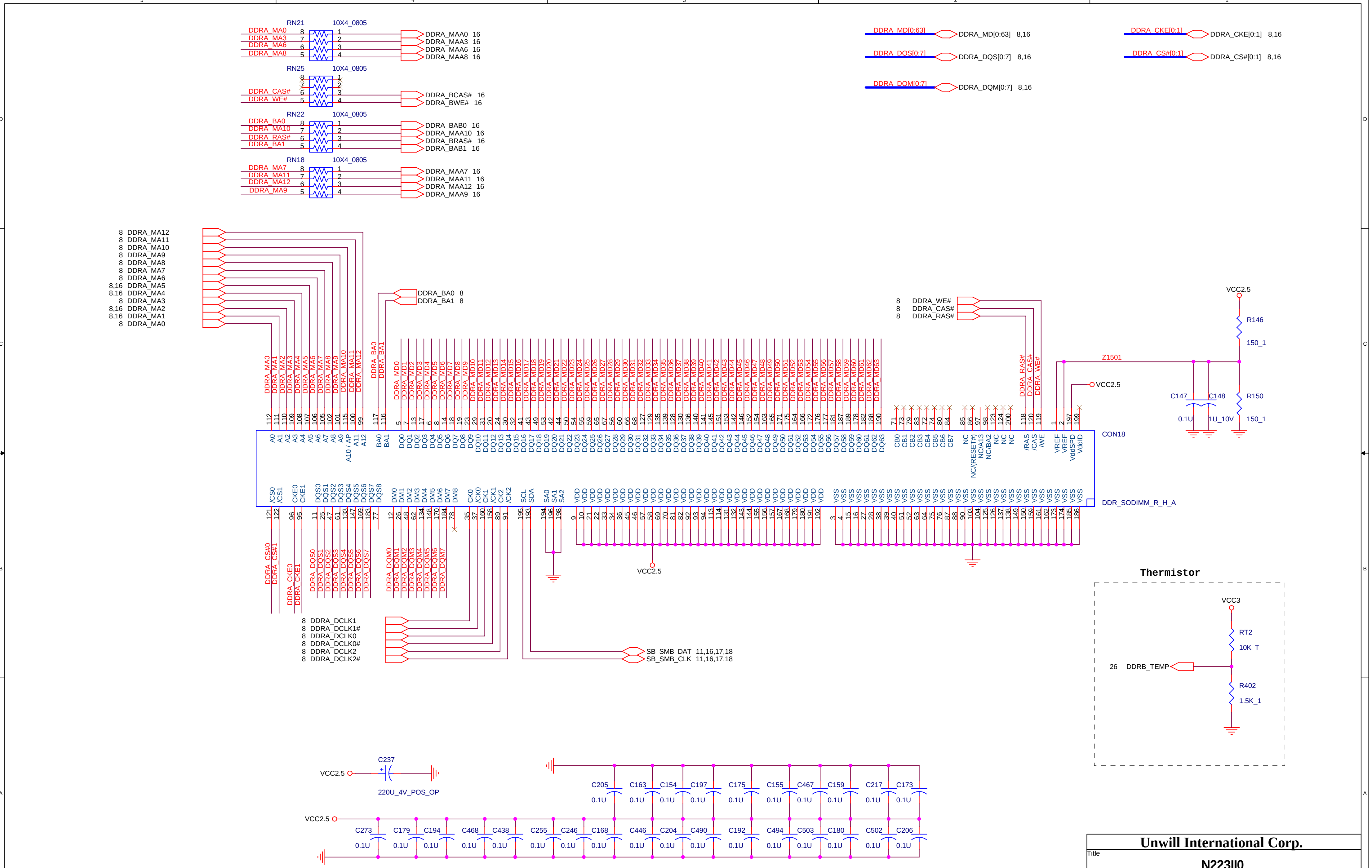


Unwill International Corp.

Title: **N223II0**

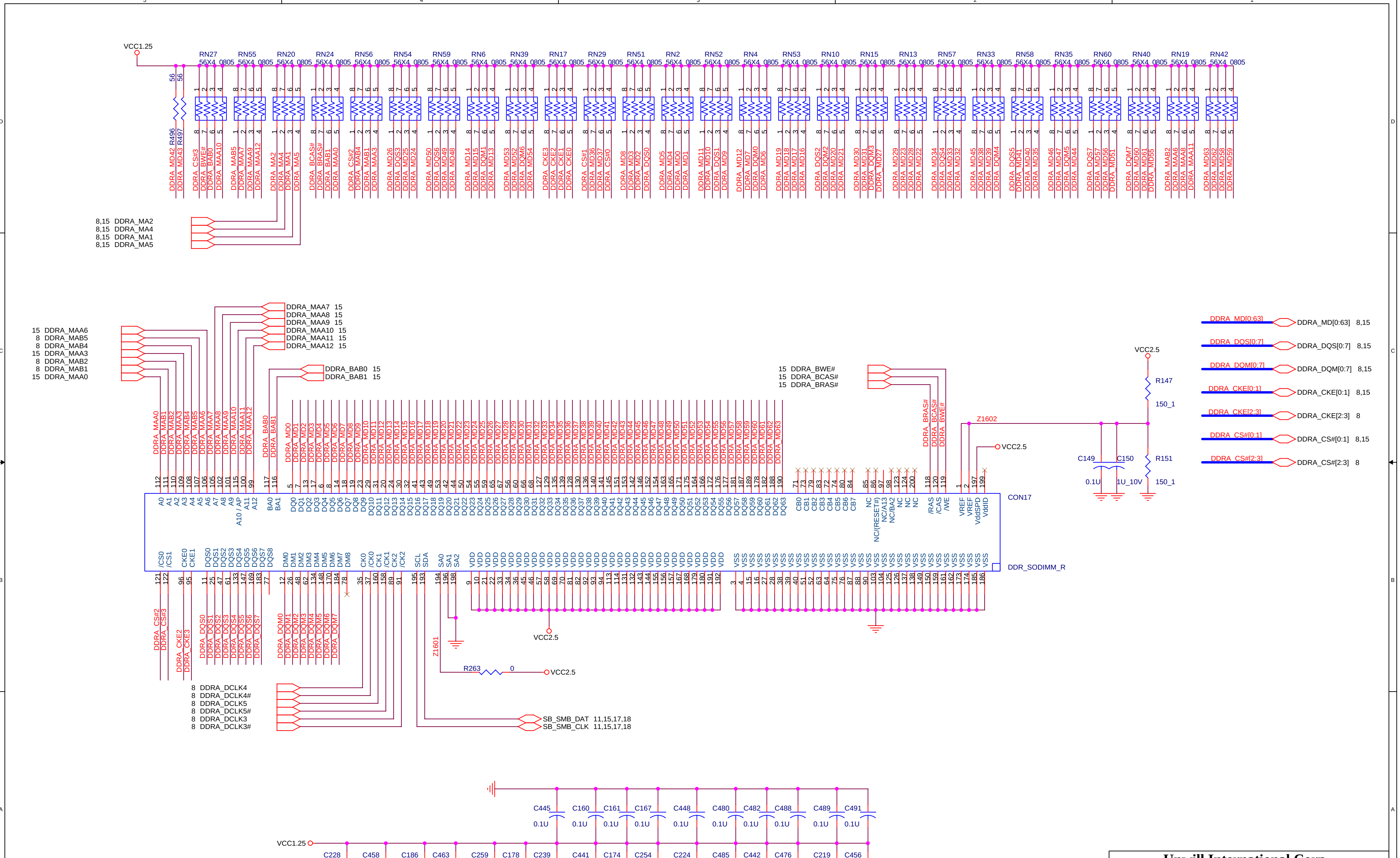
Size	Document Number		Rev
	2665	**System Pull up**	01

Date: Tuesday, May 25, 2004 | Sheet 14 of 39

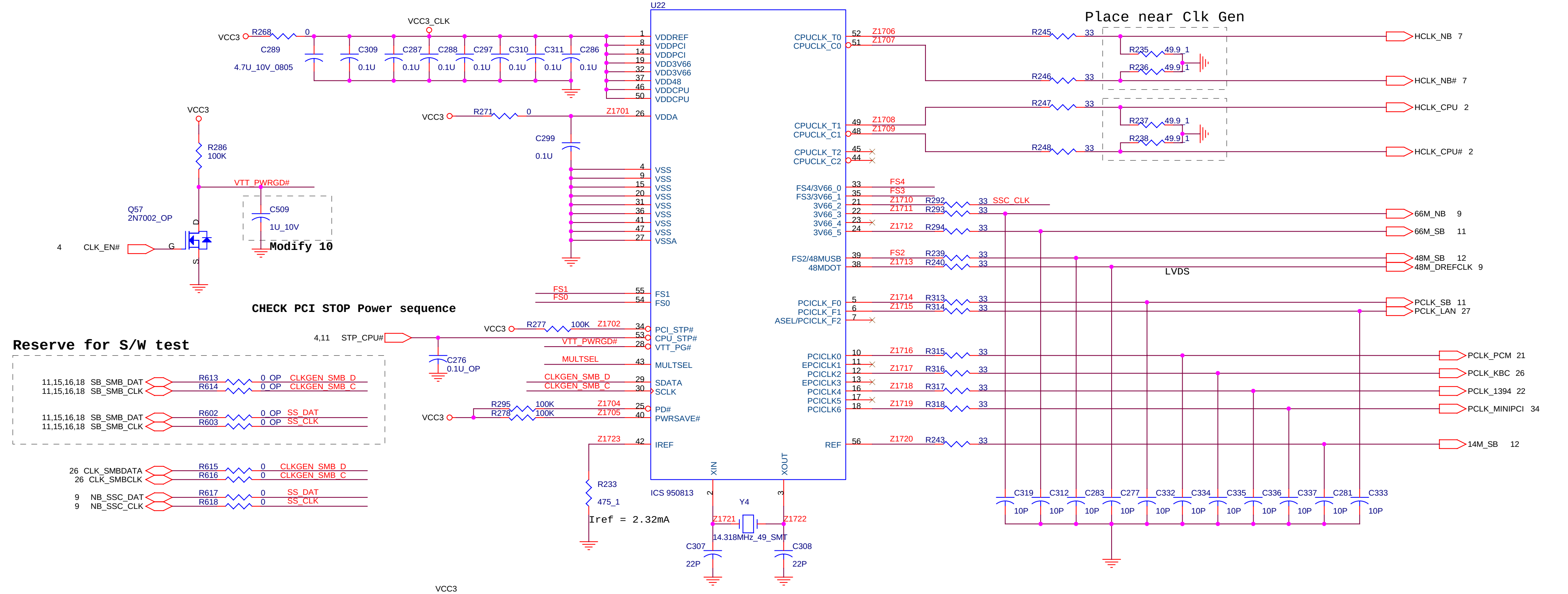


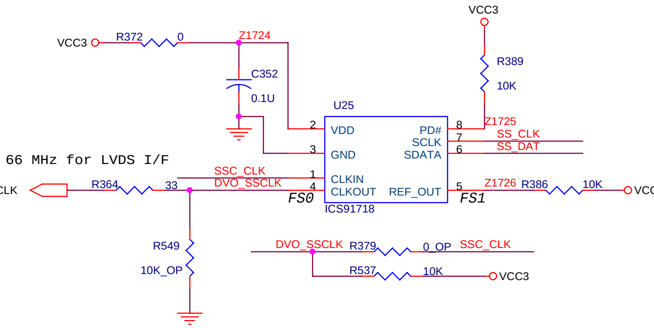
Unwill International Corp.

Title	N223II0
Size / Document Number	2665 — SODIMM -1
Rev	01
Date:	Tuesday, May 25, 2004
Sheet	15 of 39



Unwill International Corp.	
Title	N223II0
Size / Document Number	2665 — SODIMM -2
Rev	01
Date:	Tuesday, May 25, 2004
Sheet	16 of 39





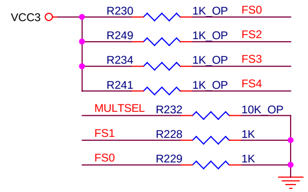
FS1	FS0	SPREAD %
1	0	-1% DOWN SPREAD
1	1	+/-1% CENTER SPREAD

FS1 : Internal pull-up resistor
FS0 : Internal pull-down resistor

Ce = 2*CL - (Cs + Ci)
CL = Crystal Load Cap = 16P
Ci = IC internal Cap = 5P
Cs = 2P
Ce = Crystal external Cap = 22P

POWER ON STRAPPING

MULTSEL internal 120K pull_up to VCC
FS2,FS3,FS4 internal 120K pull-down
to VSS



Power On Default: FSB = 100MHz, Change to
Real frequency by VCO

FS1	FS0	CPU	AGP	PCI
0	0	100	66	33
0	1	133	66	33
1	0	200	66	33
1	1	166	66	33

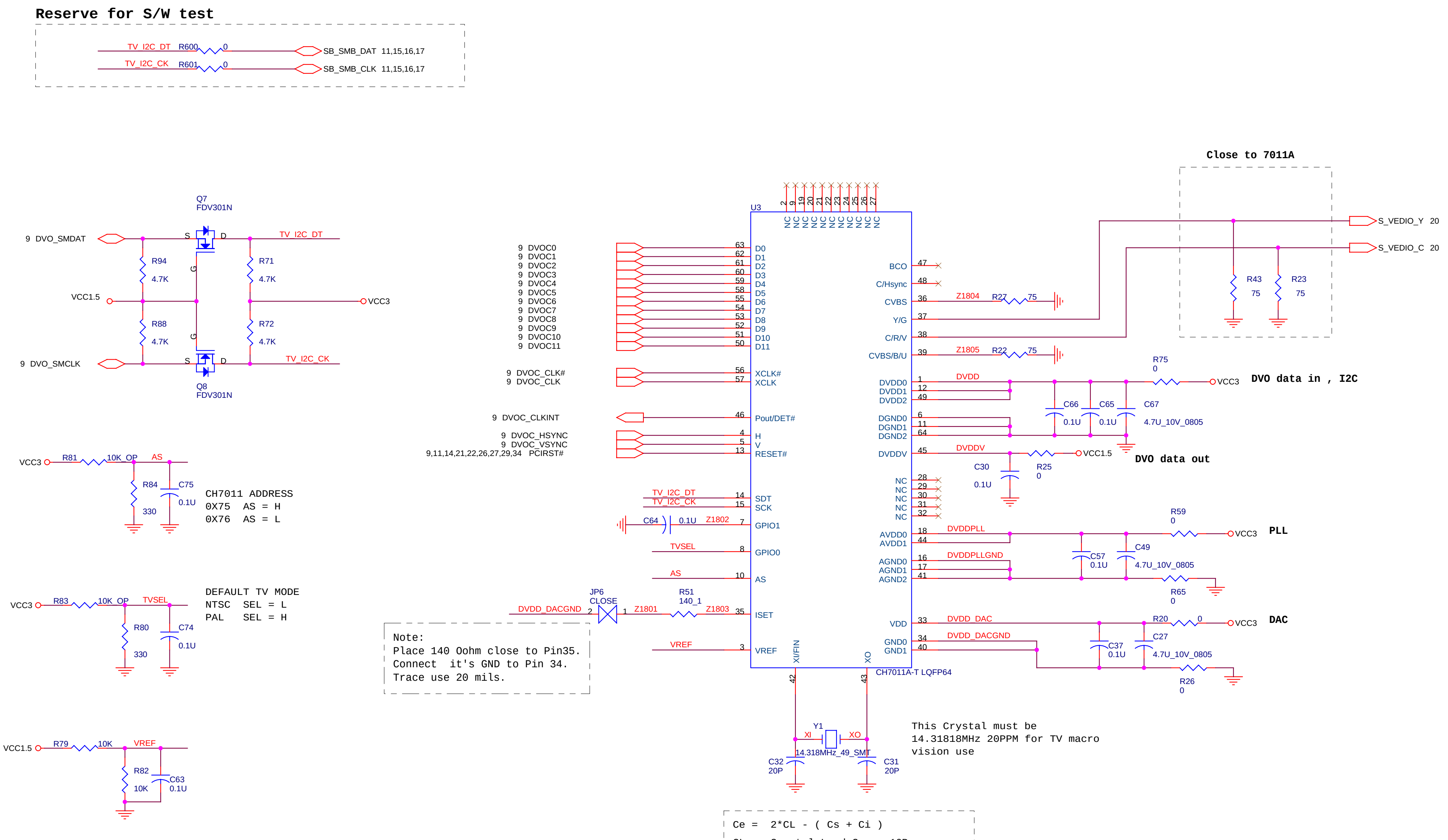
FS4	FS3	FS2	CPU, 3V66, PCI Clocking Mode
0	0	0	Standard Clocking 0.3% Center
0	0	1	Standard Clocking 0 to -0.5%, Down Spread
0	1	0	Standard Clocking 0.3% Center Spread
0	1	1	Standard Clocking 0 to - 0.75%, Down Spread
1	0	0	Pwr Save Clocking Spread Off
1	0	1	3% Overclocking 0.3% Center Spread
1	1	0	5% Overclocking 0.3% Center Spread
1	1	1	10% Overclocking 0.3% Center Spread

Unwill International Corp.

Title: N223II0

Size | Document Number: 2665 | CLK GEN ICS 950813 | Rev 01

Date: Tuesday, May 25, 2004 | Sheet 17 of 39



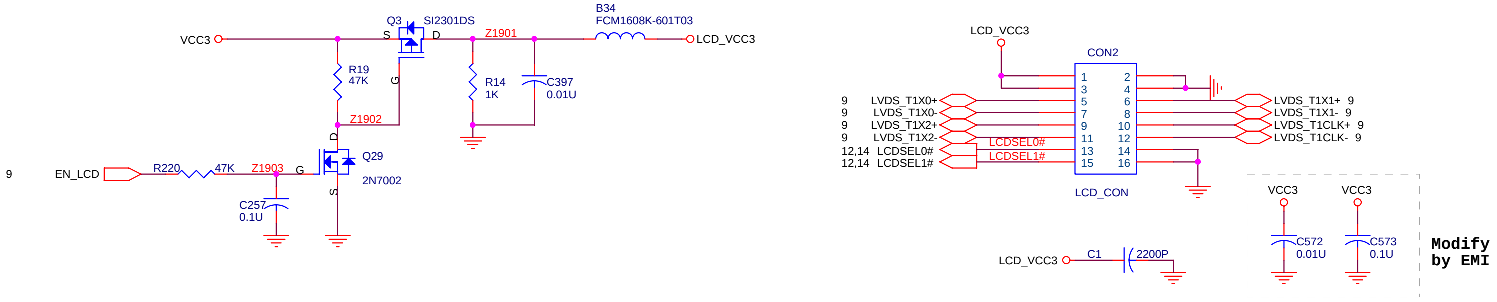
Note:
Place 140 Oohm close to Pin35.
Connect it's GND to Pin 34.
Trace use 20 mils.

This Crystal must be 14.31818MHz 20PPM for TV macro vision use

Ce = 2*CL - (Cs + Ci)
CL = Crystal Load Cap = 16P
Ci = IC internal Cap = 10 ~ 15P
Cs = 2P
Ce = Crystal external Cap = 15 ~ 20P

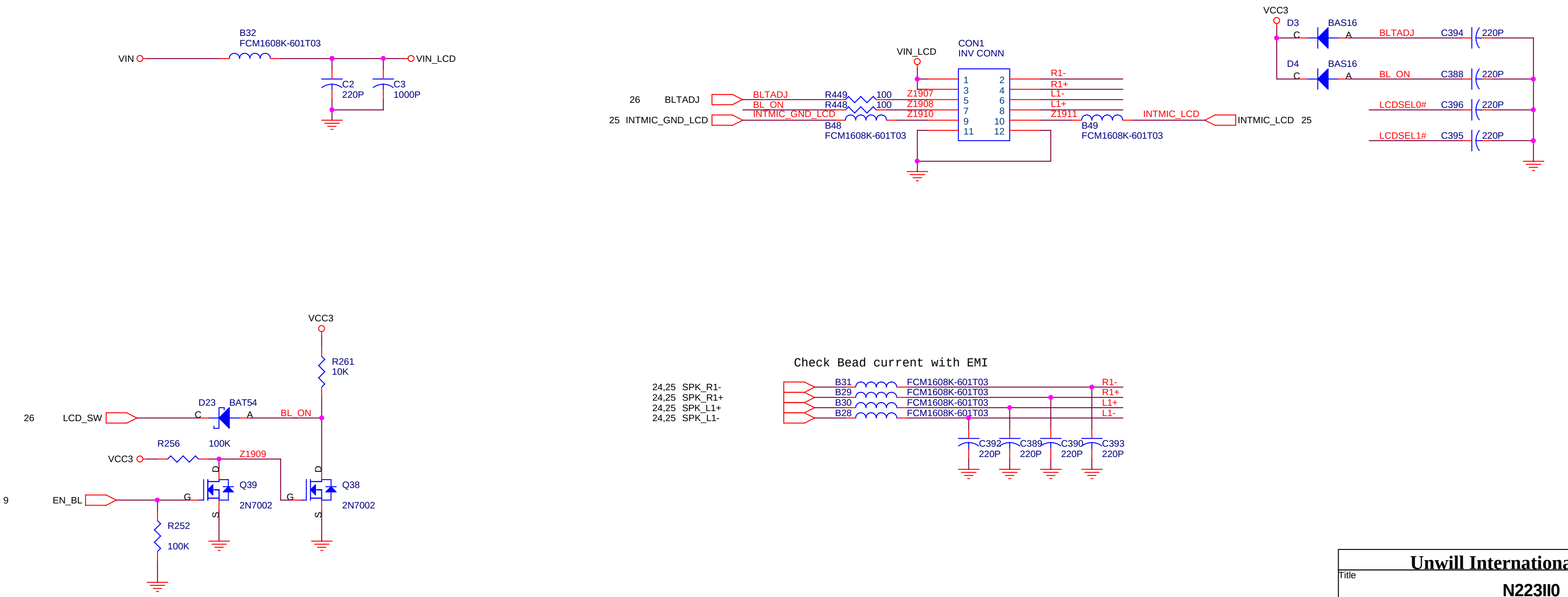
Unwill International Corp.		
Title: N223II0		
Size	Document Number: 2665 DVO CH7011 TV	Rev: 01
Date: Tuesday, May 25, 2004	Sheet 18 of 39	

LCD I/F



LCD TYPE	18Bit WXGA	18Bit XGA	RESERVE	RESERVE
LCDSEL1	1	1	0	0
LCDSEL0	0	1	0	1

INVERTOR I/F

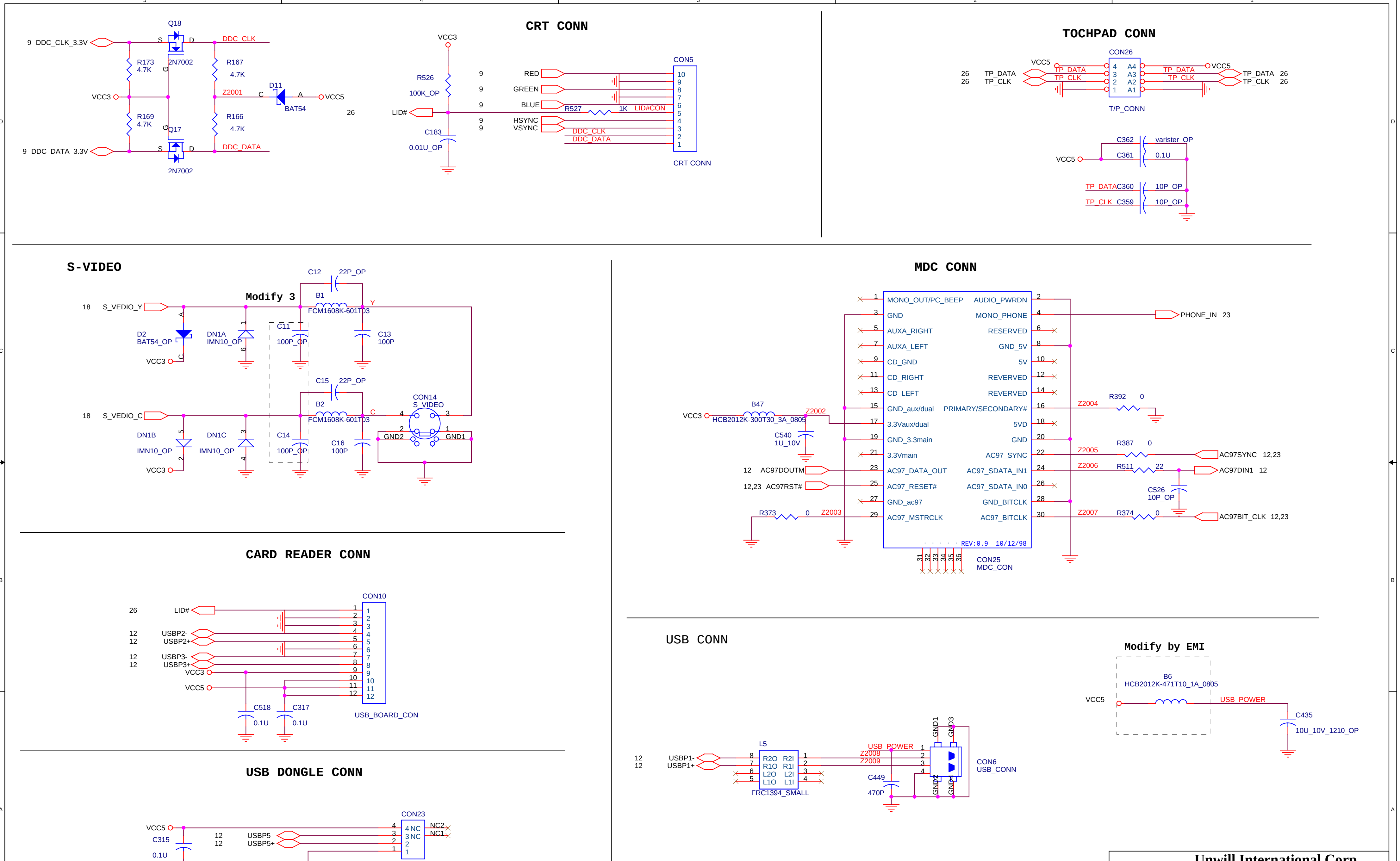


Unwill International Corp.

Title: N223II0

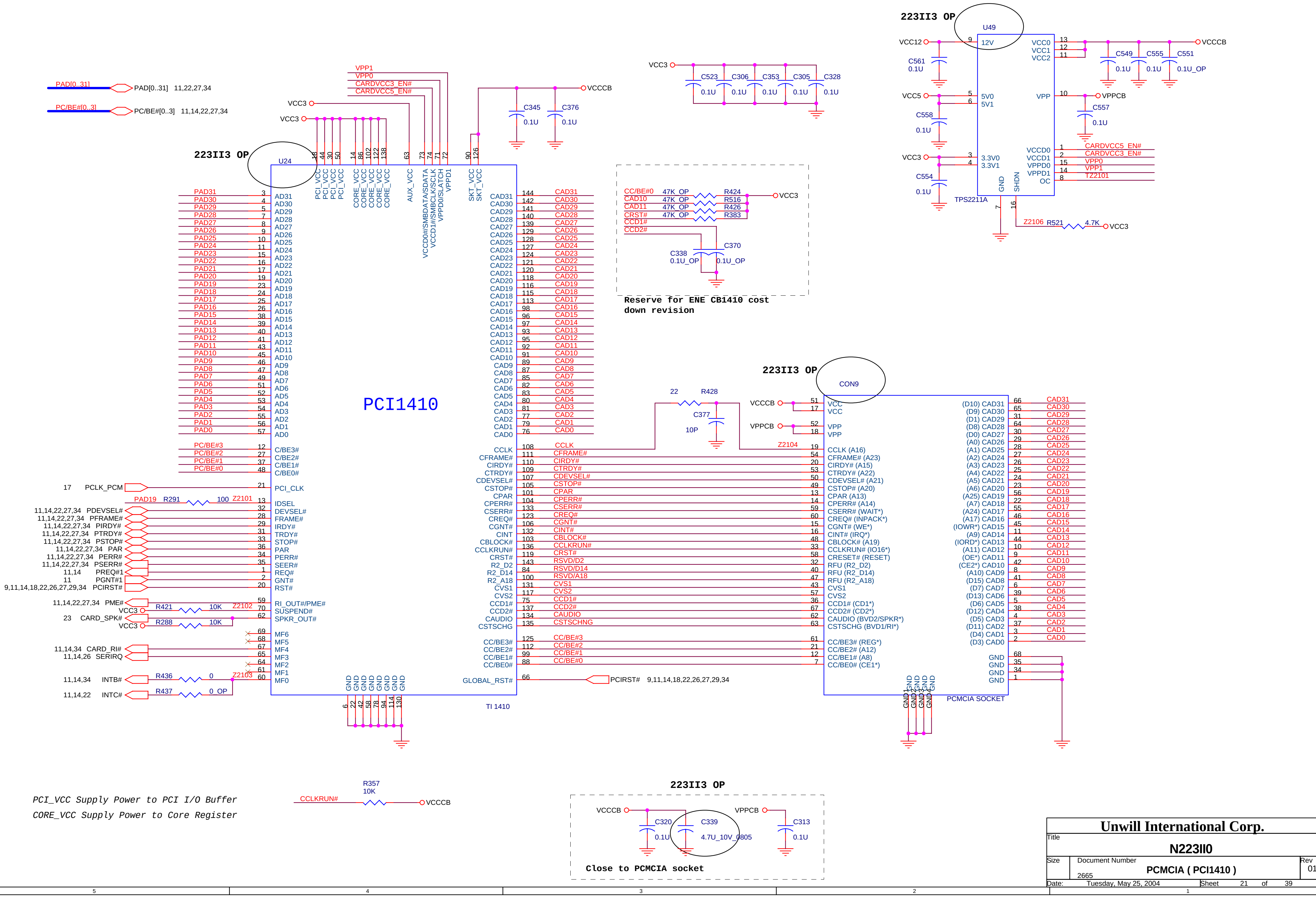
Size | Document Number: 2665 | LCD CONNECTOR | Rev 01

Date: Tuesday, May 25, 2004 | Sheet 19 of 39



Unwill International Corp.

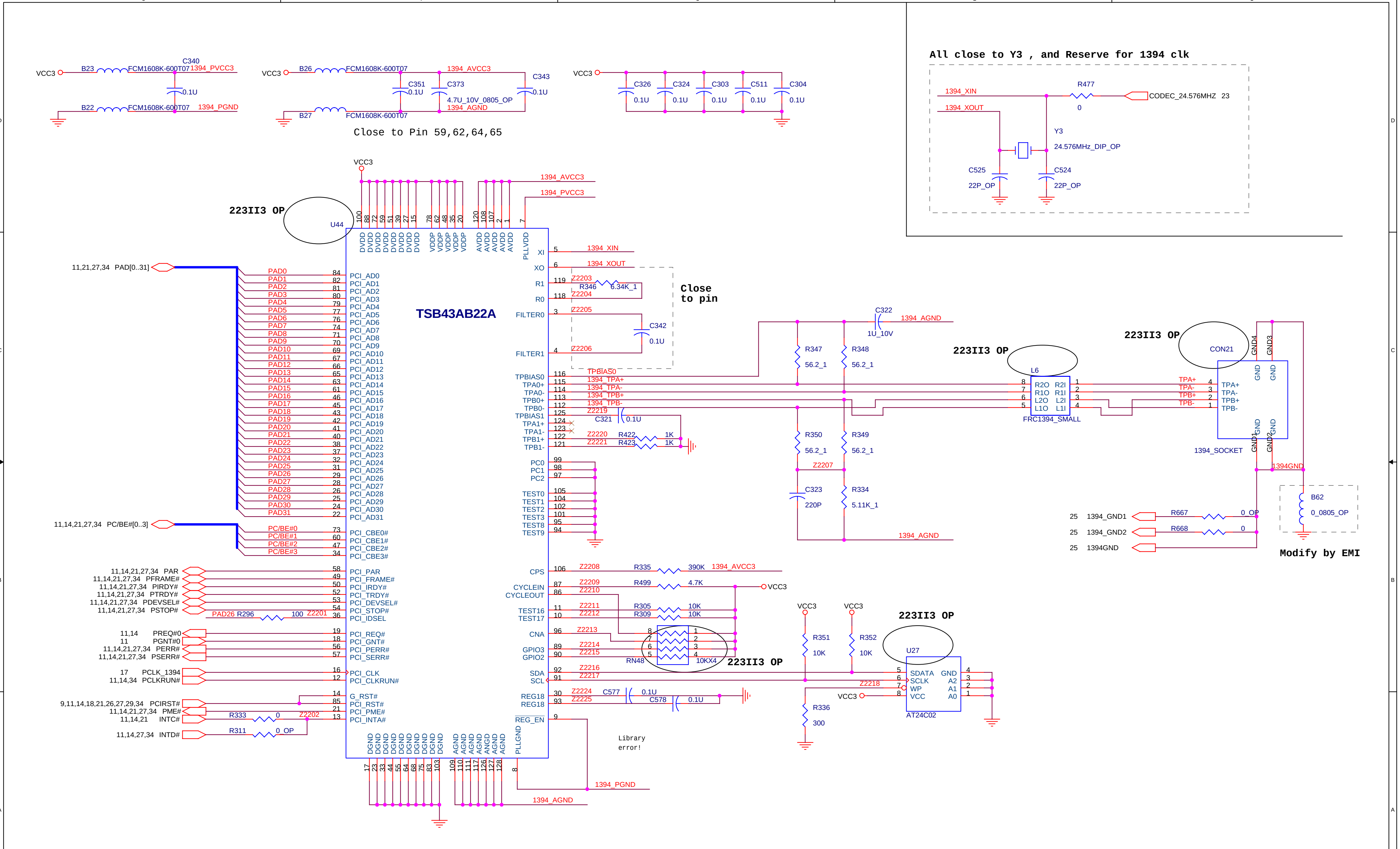
Title	N223II0
Size / Document Number	2665
	I/O CONNECTORS
Rev	01
Date:	Tuesday, May 25, 2004
Sheet	20 of 39



PCI_VCC Supply Power to PCI I/O Buffer

CORE_VCC Supply Power to Core Register

Unwill International Corp.	
Title	N223II0
Size / Document Number	2665 PCMCIA (PCI1410)
Rev	01
Date:	Tuesday, May 25, 2004
Sheet	21 of 39

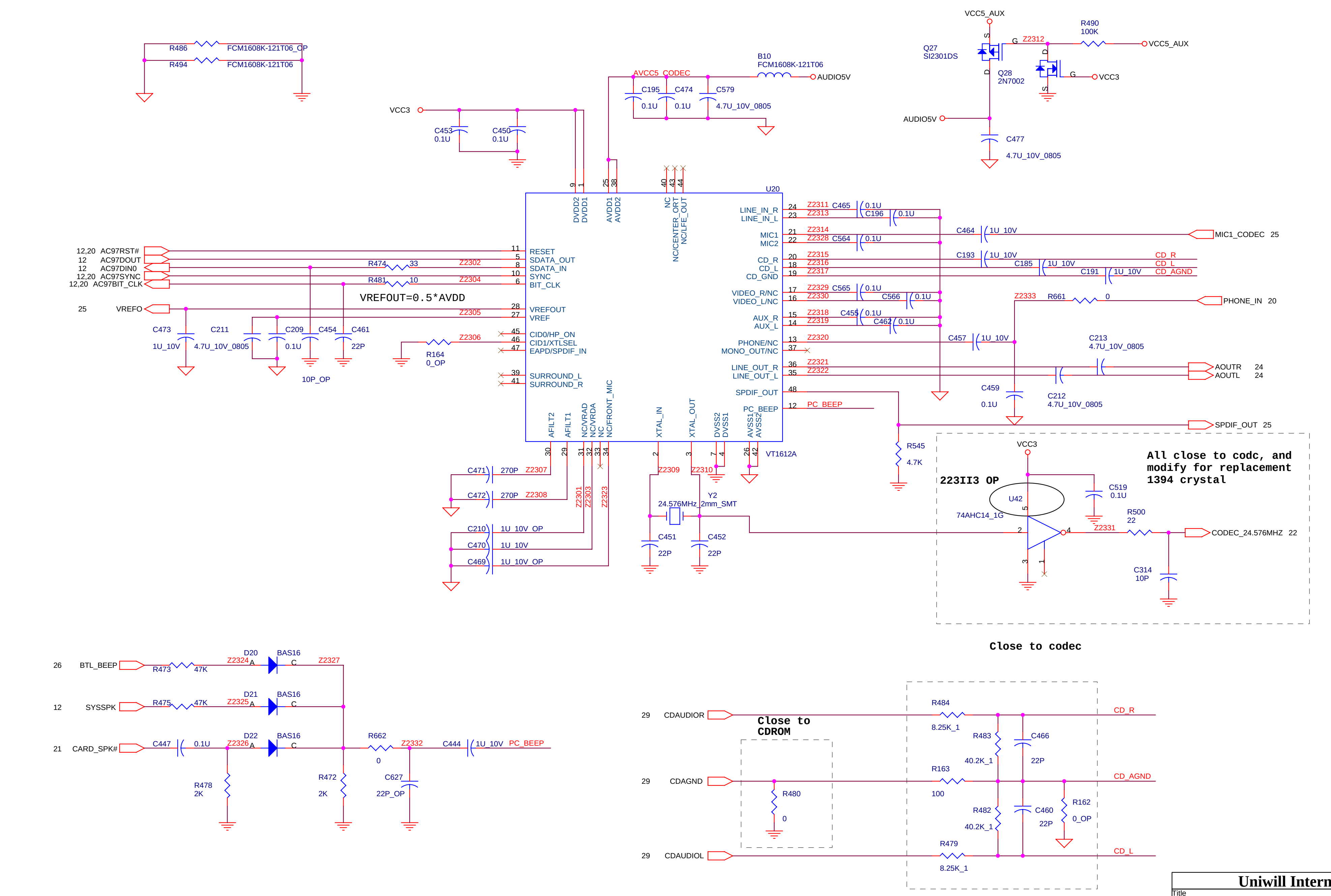


Uniwill International Corp.

Title: N223II0

Size | Document Number: 2665 | IEEE 1394 (TSB43AB22A) | Rev 01

Date: Tuesday, May 25, 2004 | Sheet 22 of 39

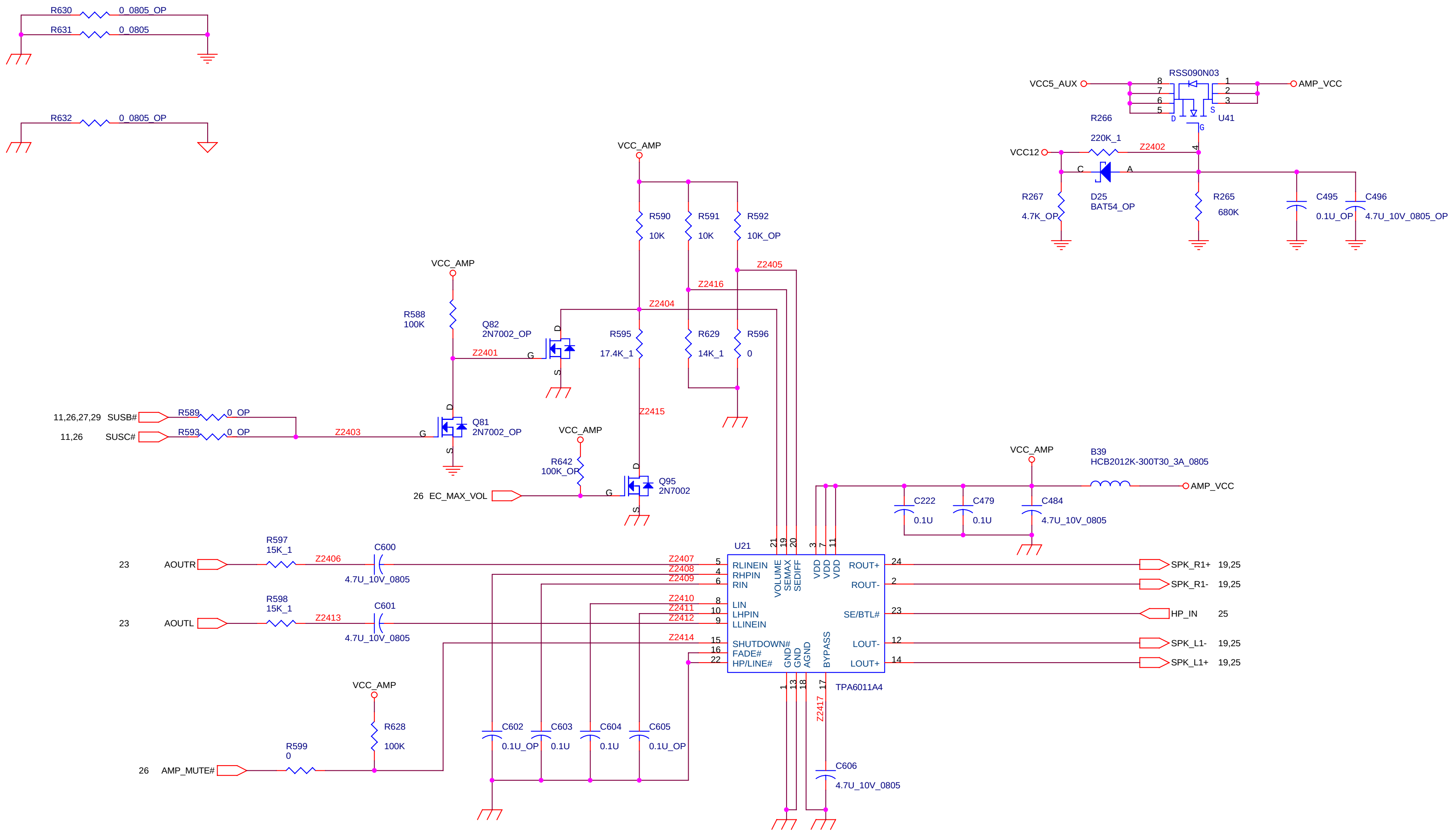


Uniwill International Corp.

Title: N223II0

Size	Document Number	Rev
	2665 CODEC-VT1612A	01

Date: Tuesday, May 25, 2004 — Sheet 23 of 39

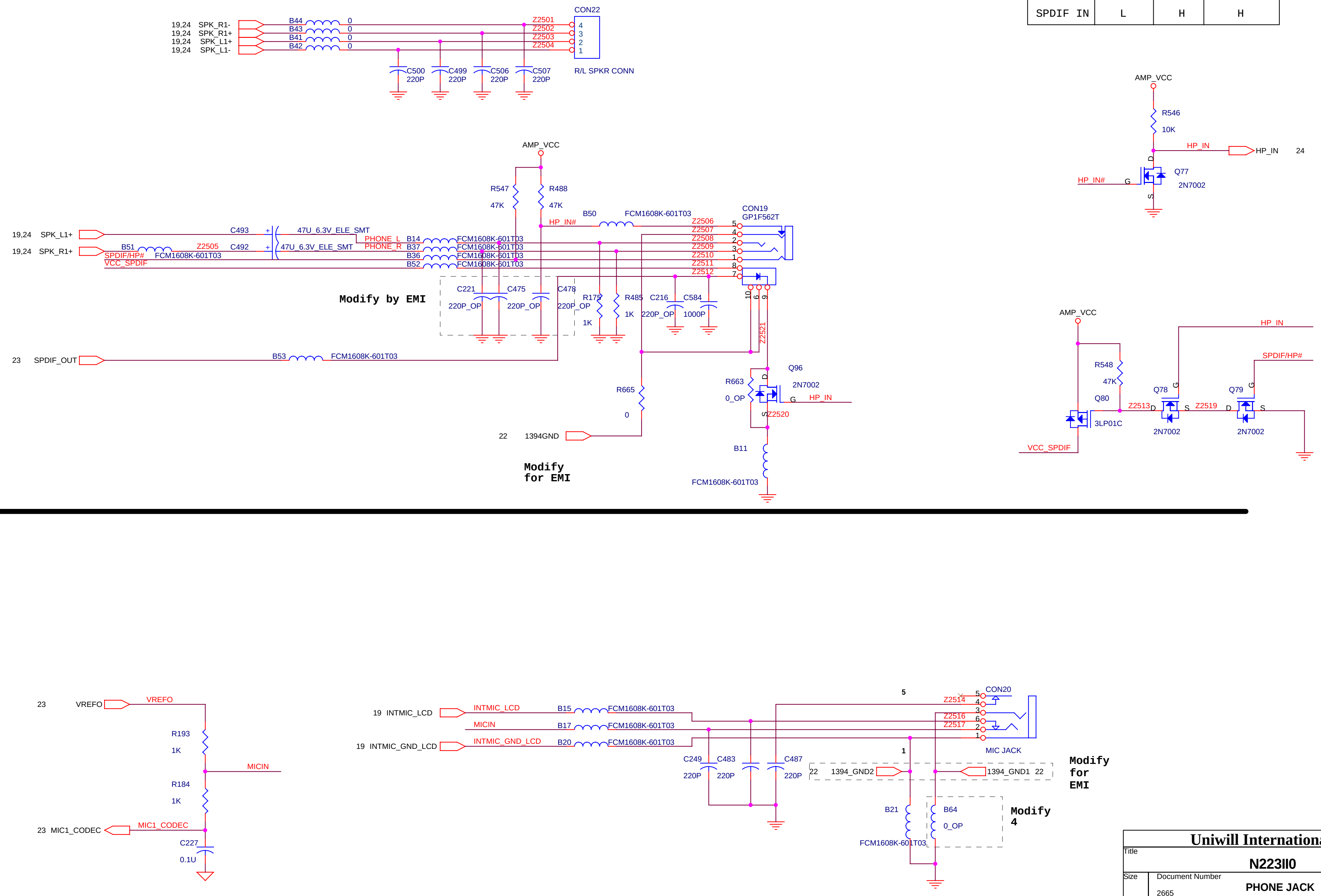


Uniwill International Corp.

Title	N223II0	
Size	Document Number: AMP TI-TPA6011A4 (2665)	Rev 01
Date: Tuesday, May 25, 2004	Sheet 24 of 39	

Check Bead current with EMI

	HP_IN#	HP_IN	SPDIF/HP#
H.J IN	L	H	L
SPDIF IN	L	H	H



Uniwill International Corp.

Title	N223II0
Size / Document Number	2665 PHONE JACK
Rev	01
Date:	Tuesday, May 25, 2004
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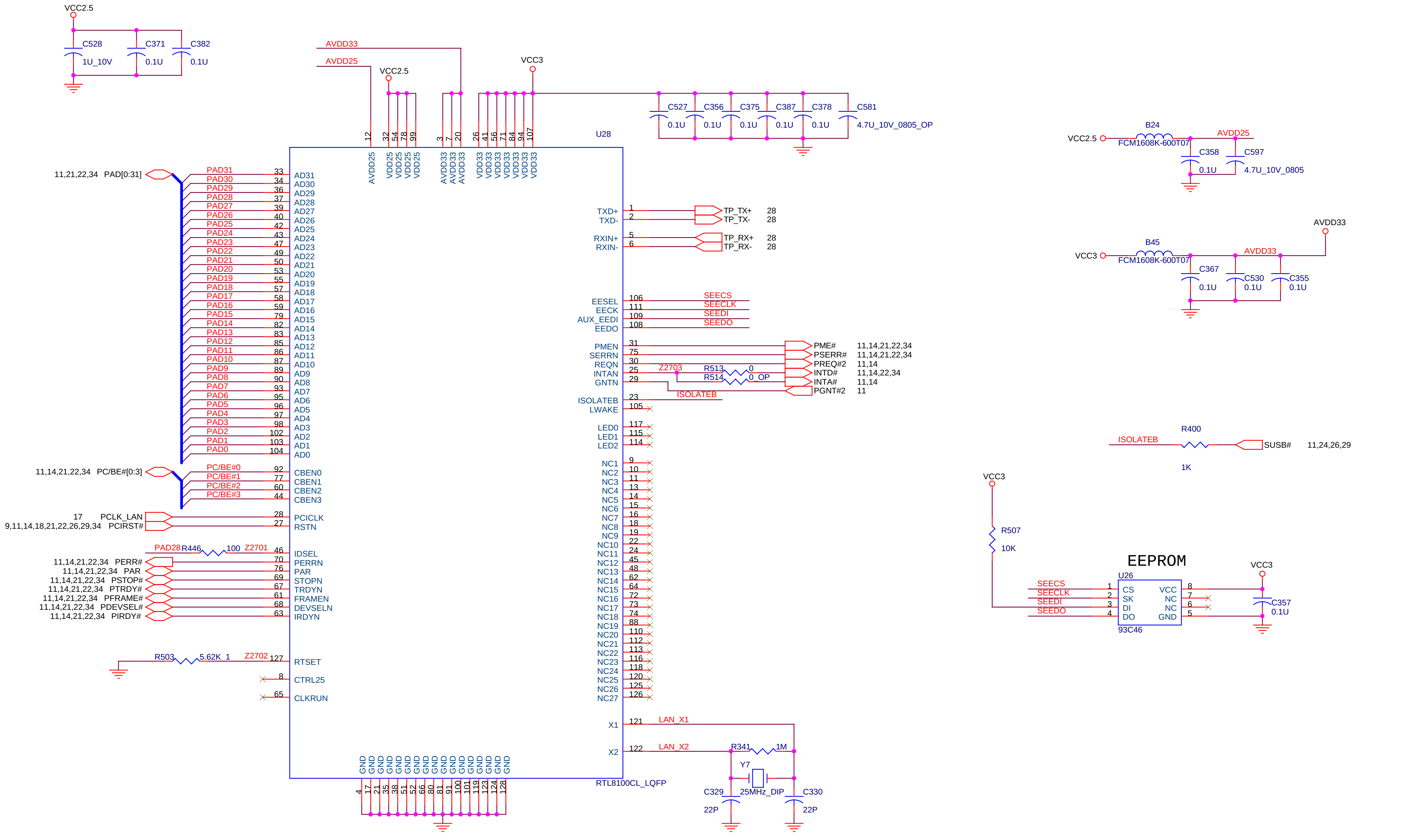
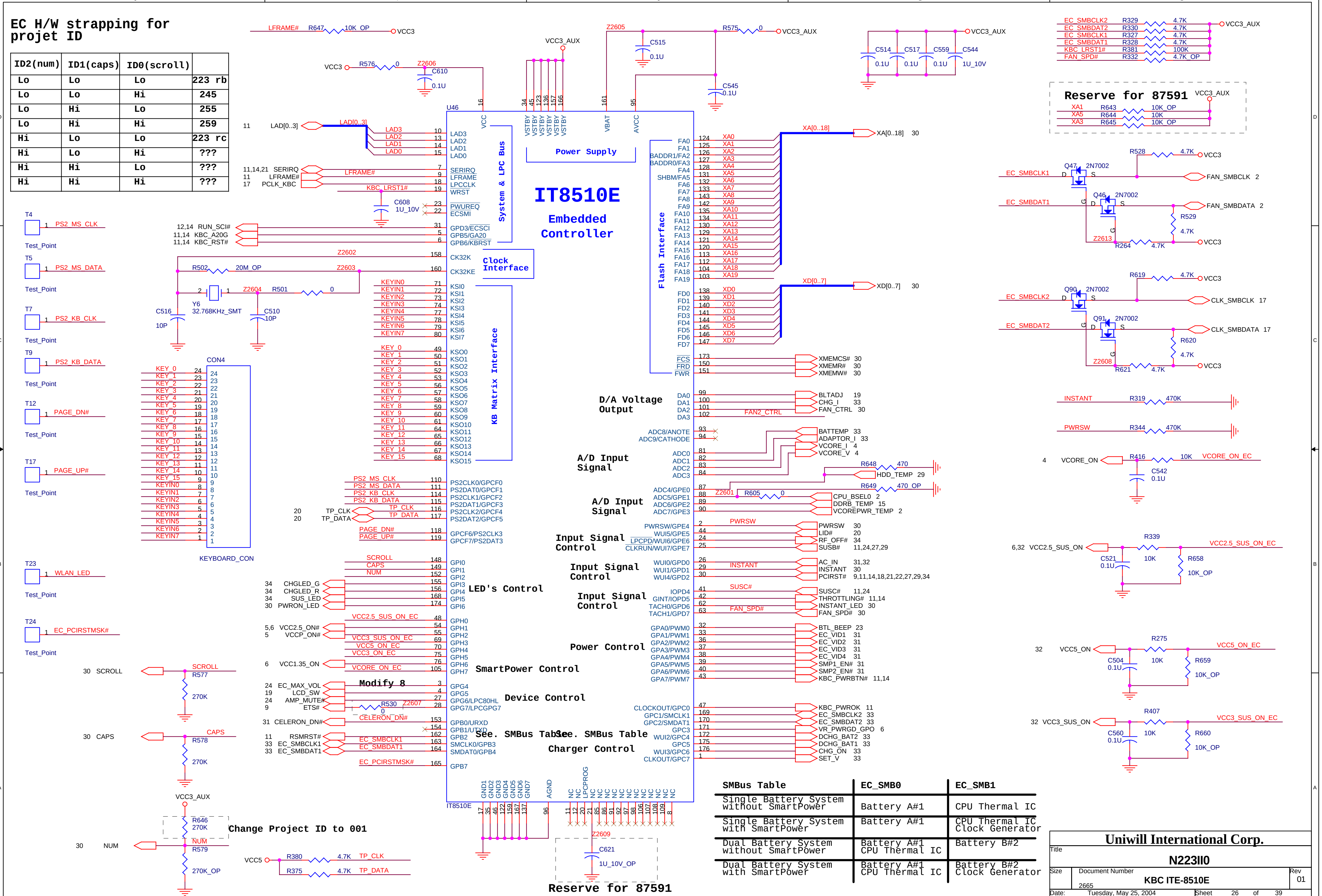


Uniwill International Corp.

Title	N223II0	
Size	Document Number: 2665 LAN - RTL8100CL	Rev 01
Date: Tuesday, May 25, 2004	Sheet 27 of 39	

EC H/W strapping for projet ID

ID2(num)	ID1(caps)	ID0(scroll)	
Lo	Lo	Lo	223 rb
Lo	Lo	Hi	245
Lo	Hi	Lo	255
Lo	Hi	Hi	259
Hi	Lo	Lo	223 rc
Hi	Lo	Hi	???
Hi	Hi	Lo	???
Hi	Hi	Hi	???



SMBus Table

	EC_SMB0	EC_SMB1
Single Battery System without SmartPower	Battery A#1	CPU Thermal IC
Single Battery System with SmartPower	Battery A#1	CPU Thermal IC Clock Generator
Dual Battery System without SmartPower	Battery A#1 CPU Thermal IC	Battery B#2
Dual Battery System with SmartPower	Battery A#1 CPU Thermal IC	Battery B#2 Clock Generator

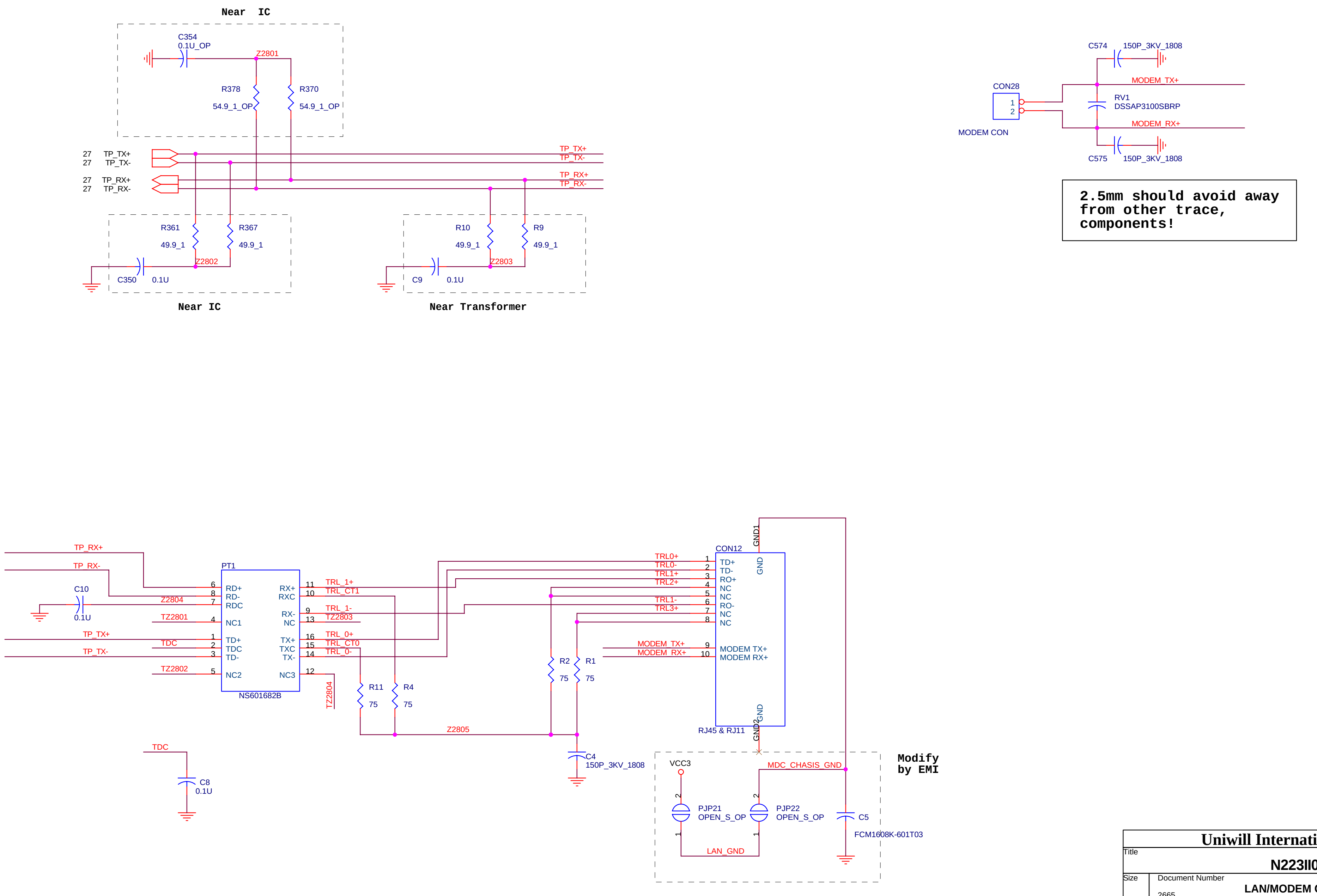
Uniwill International Corp.

Title: N223II0

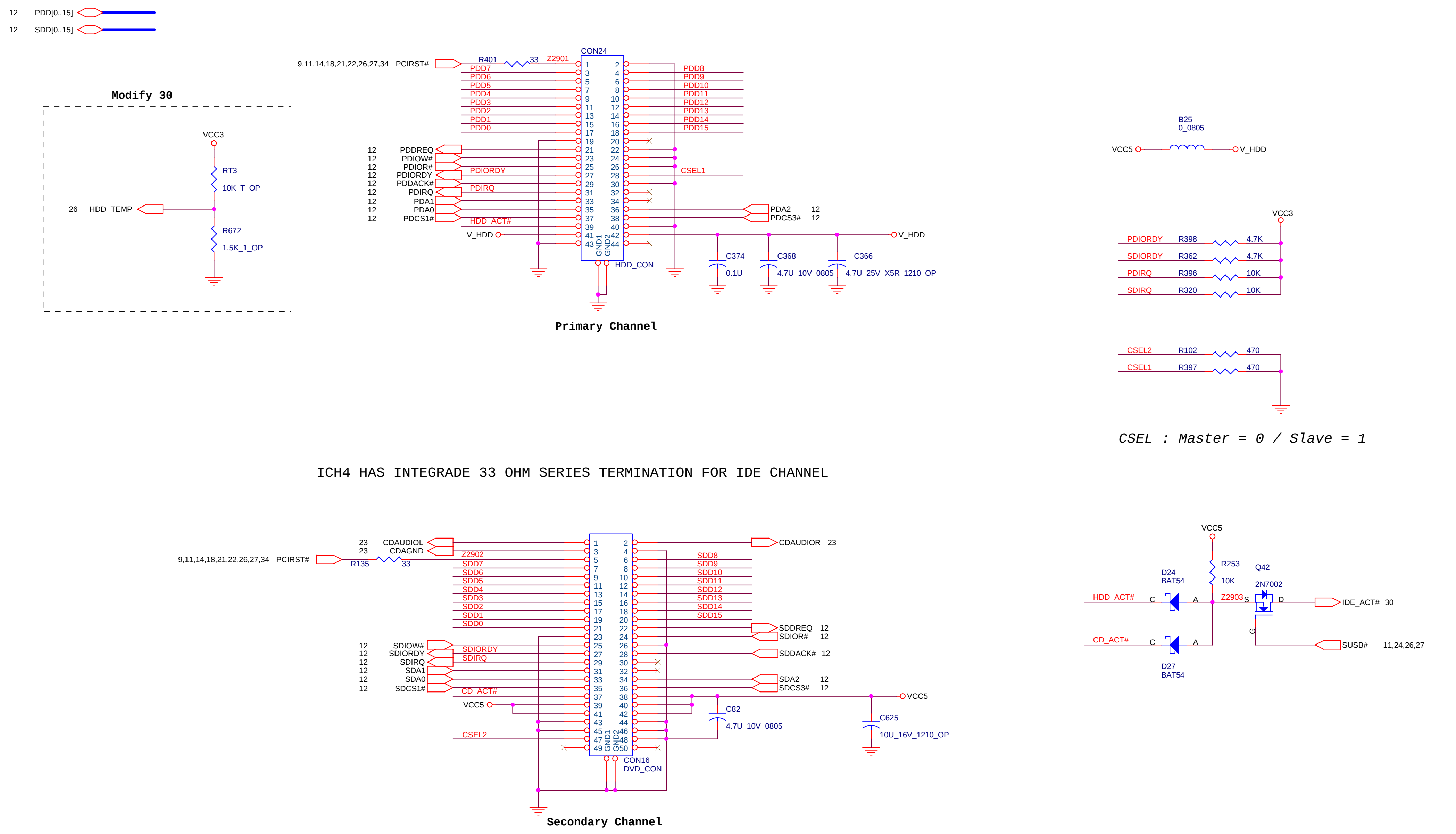
Size | Document Number: KBC ITE-8510E | Rev: 01

2665

Date: Tuesday, May 25, 2004 | Sheet 26 of 39



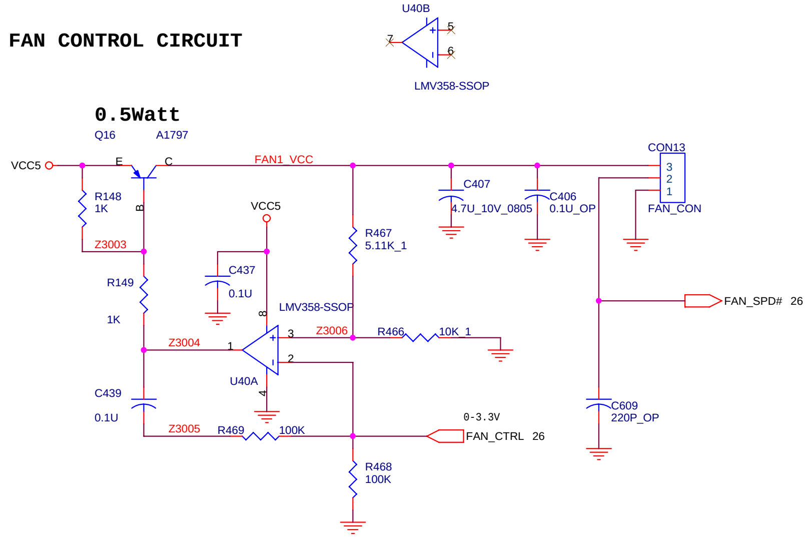
Uniwill International Corp.			
Title	N223II0		
Size	Document Number 2665	LAN/MODEM CONNECTOR	Rev 01
Date: Tuesday, May 25, 2004		Sheet 28 of 39	

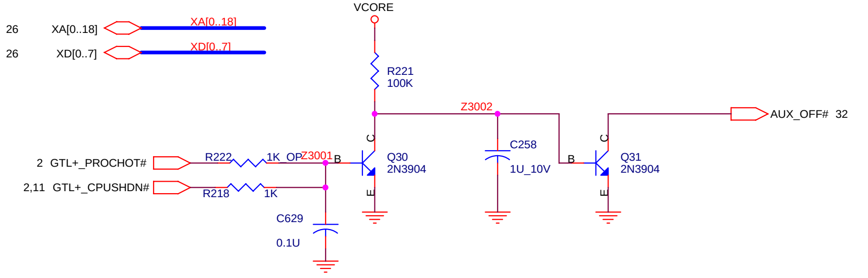


Uniwill International Corp.		
Title	N223II0	
Size	Document Number 2665 — HDD & CDROM	Rev 01
Date: Tuesday, May 25, 2004	Sheet 29 of 39	

FAN CONTROL CIRCUIT

0.5Watt





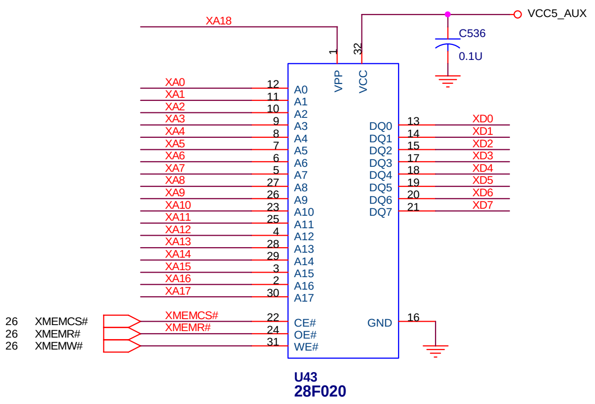
CPU HOT THROTTLING/SHUT DOWN PROTECT CIRCUIT

THERMAL SOLUTION 1:
USE CPUHOT# TO IMPLEMENT H/W THROTTLING AND SHUT DOWN.
EC READ TEMPRATURE AND CONTROL FAN.

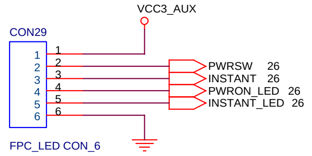
THERMAL SOLUTION 2:
USE 1617 TO READ TEMPRATURE AND EC CONTROL FAN, THROTTLING AND SHUT DOWN.
CPU SHUT DOWN PIN IS FOR H/W PROTECT

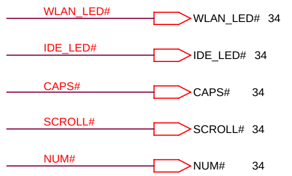
LED FPC

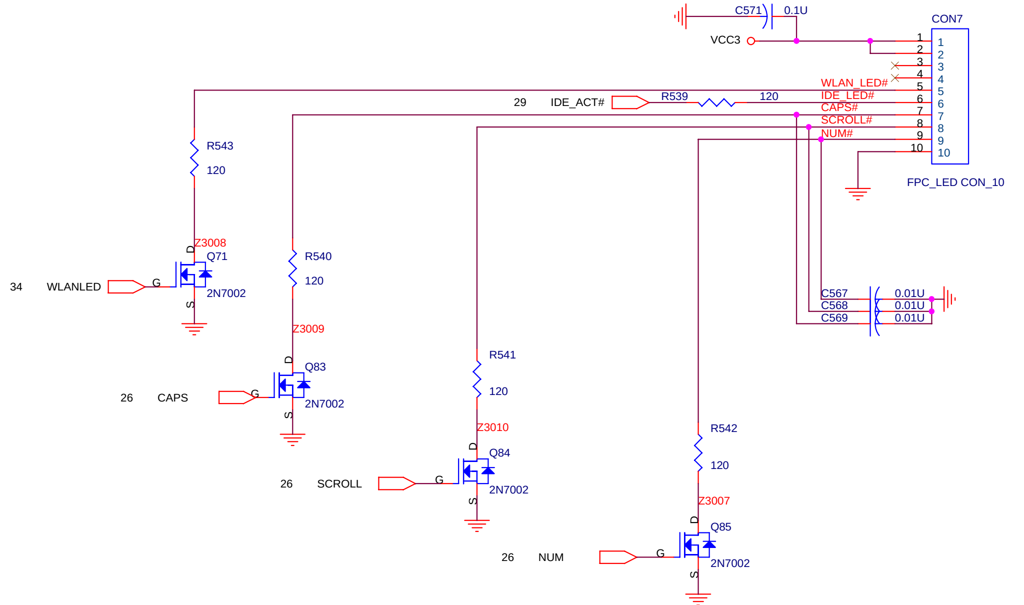
BIOS



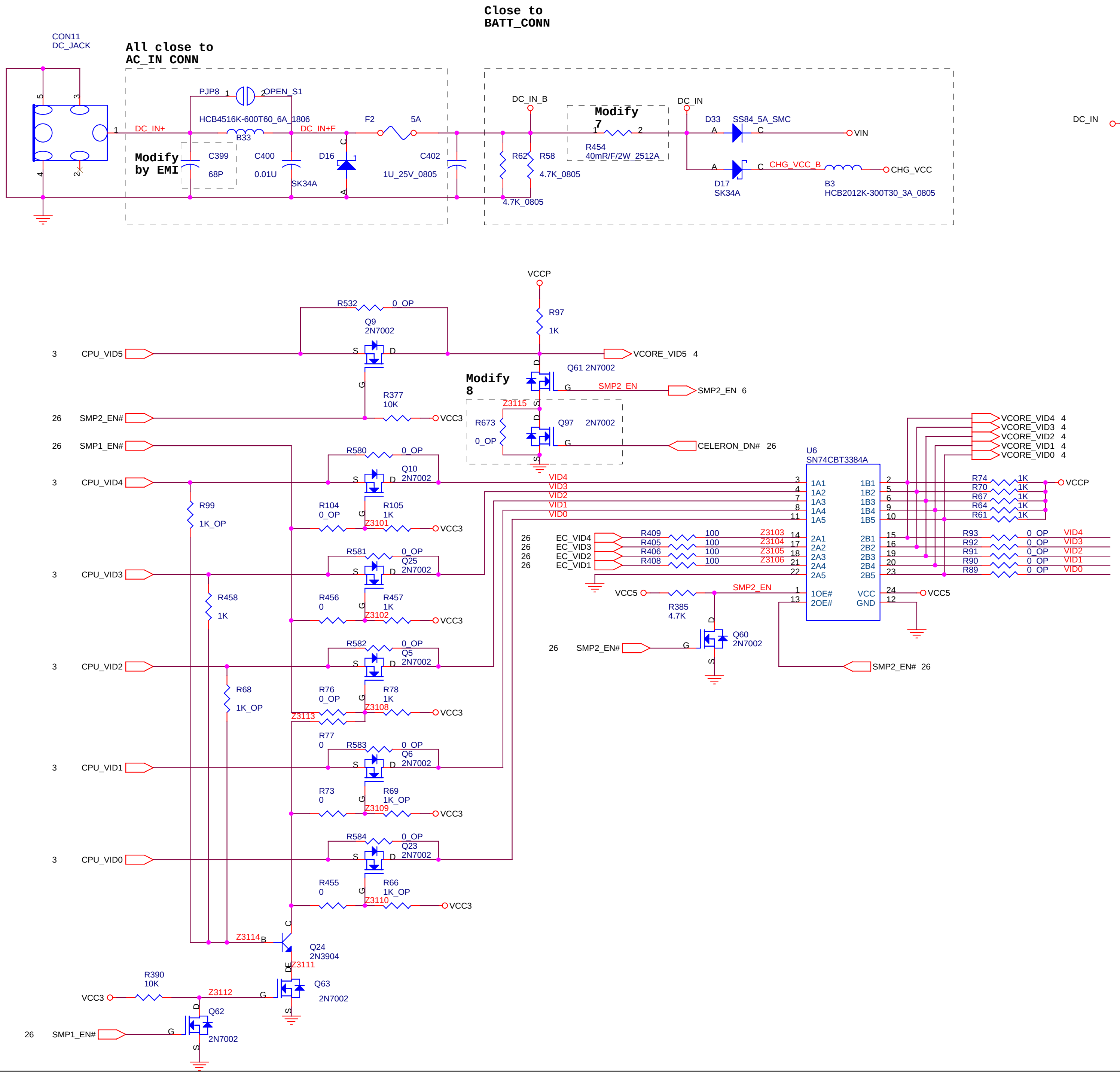
SWITCH







Uniwill International Corp.	
Title	N223II0
Size / Document Number: 2665	FAN & SHDN& LED CON & PWR — Rev 01
Date: Tuesday, May 25, 2004	Sheet 30 of 39



VID5	VID4	VID3	VID2	VID1	VID0	VCORE	+_mV
0	0	0	0	0	0	1.708	-0mV
0	0	0	0	1	0	1.676	-32mV
0	0	0	1	0	0	1.644	-64mV
0	0	1	0	0	0	1.580	-128mV
0	1	0	0	0	0	1.452	-256mV
0	0	1	0	1	0	1.548	
0	0	1	0	1	1	1.532	
0	0	1	1	0	0	1.516	
0	0	1	1	1	0	1.484	
0	1	0	0	0	1	1.436	
0	1	0	0	1	0	1.420	
0	1	0	1	0	0	1.388	
0	1	0	1	1	0	1.356	
0	1	1	0	0	1	1.308	
0	1	1	0	1	0	1.292	
0	1	1	0	1	1	1.276	
0	1	1	1	0	0	1.260	
0	1	1	1	1	0	1.228	
1	0	0	0	0	0	1.196	
1	0	0	0	1	0	1.164	
1	0	0	1	0	1	1.116	
1	0	0	1	1	0	1.100	
1	0	1	0	1	0	1.036	
1	0	1	0	1	1	1.020	
1	0	1	1	0	0	1.004	
1	0	1	1	0	1	0.988	
1	0	1	1	1	0	0.972	
1	0	1	1	1	1	0.956	
1	1	0	1	1	0	0.844	
1	1	1	1	1	1	0.716	

Uniwill International Corp.

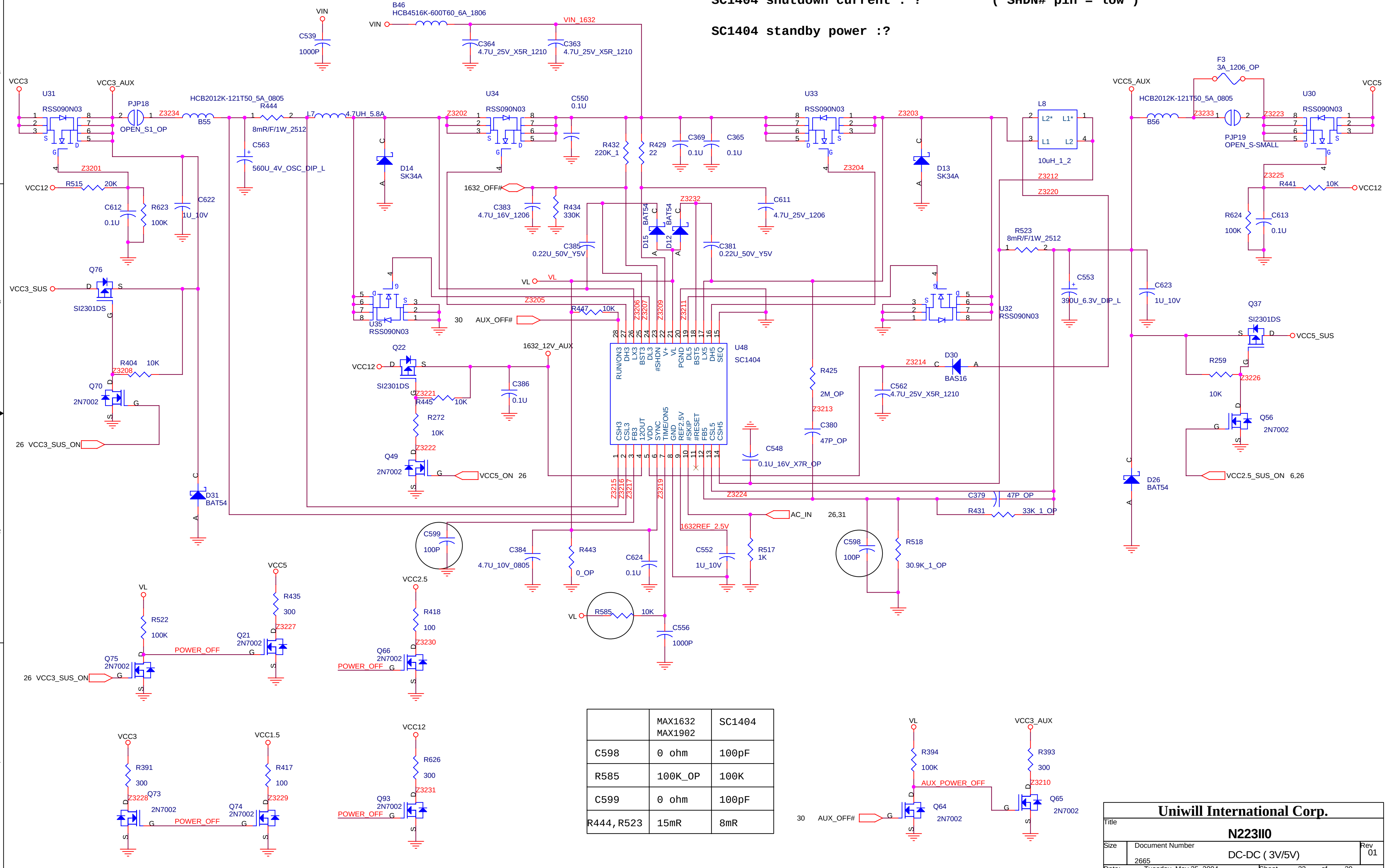
Title: N223II0

Size | Document Number: 2665 | ADAPTER & LED | Rev: 01

Date: Tuesday, May 25, 2004 | Sheet 31 of 39

SC1404 shutdown current : ?   (SHDN# pin = low)

SC1404 standby power :?



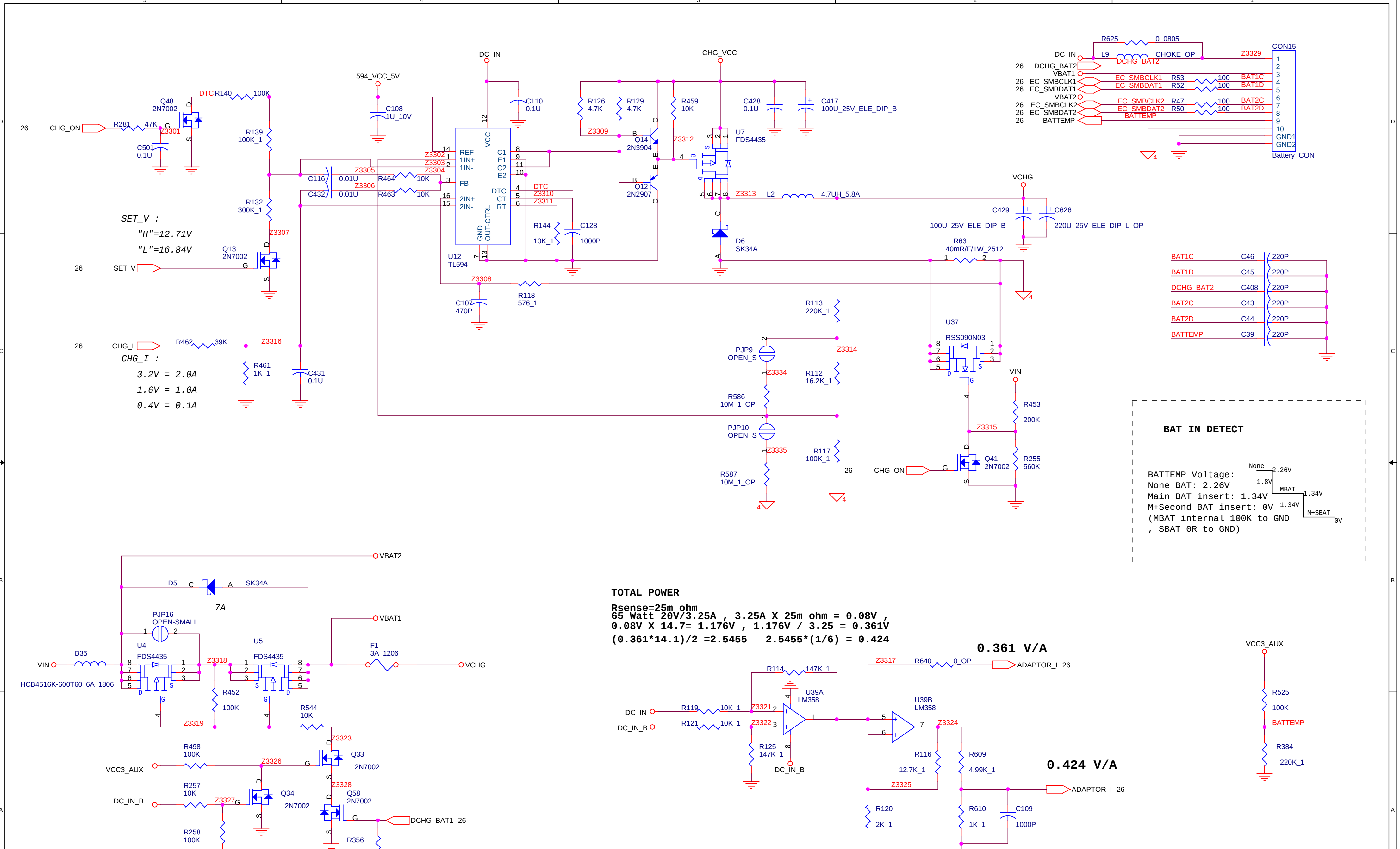
	MAX1632 MAX1902	SC1404
C598	0 ohm	100pF
R585	100K_OP	100K
C599	0 ohm	100pF
R444,R523	15mR	8mR

Uniwill International Corp.

Title: N223II0

Size | Document Number: 2665 | DC-DC (3V/5V) | Rev 01

Date: Tuesday, May 25, 2004 | Sheet 32 of 39

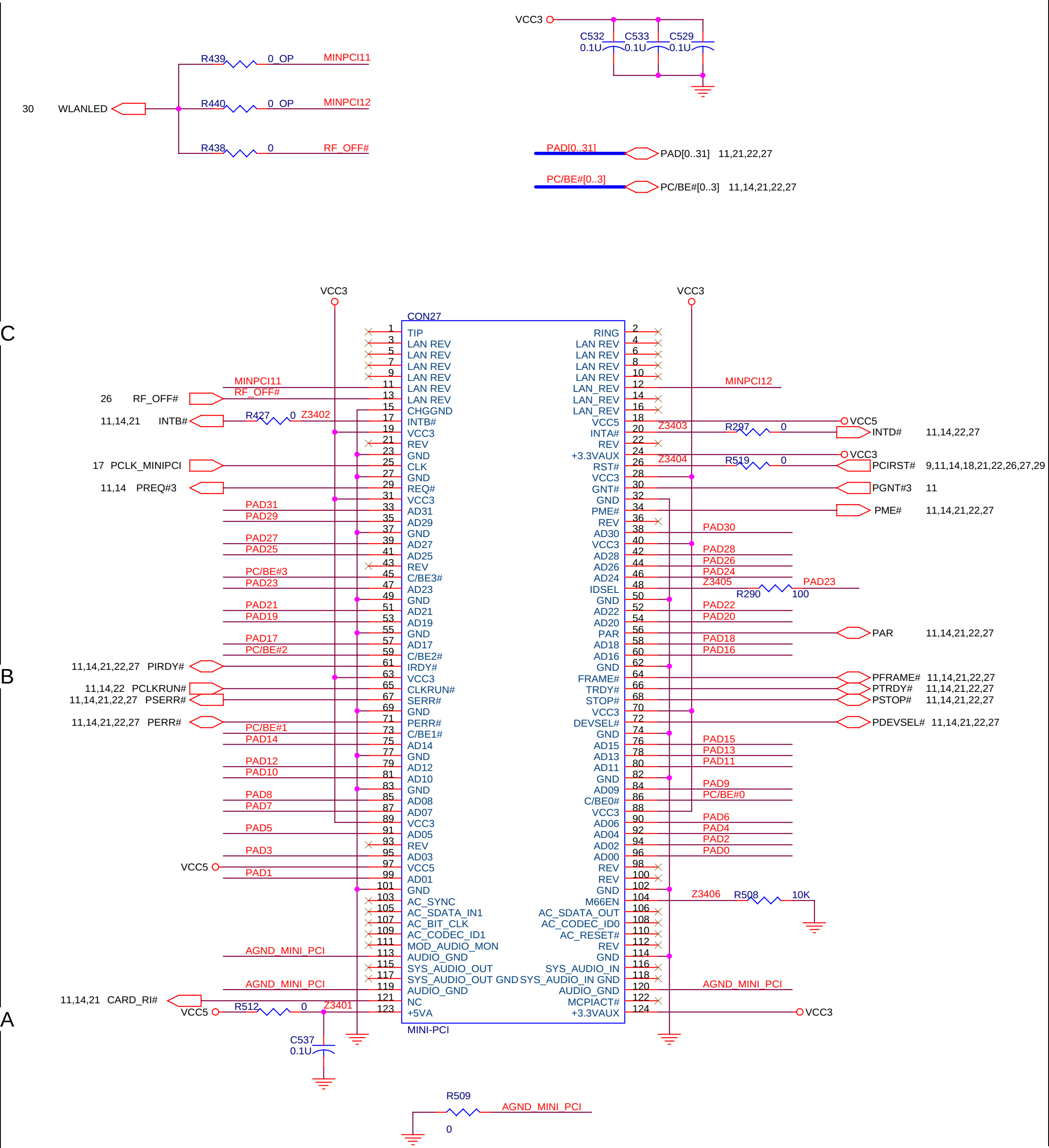


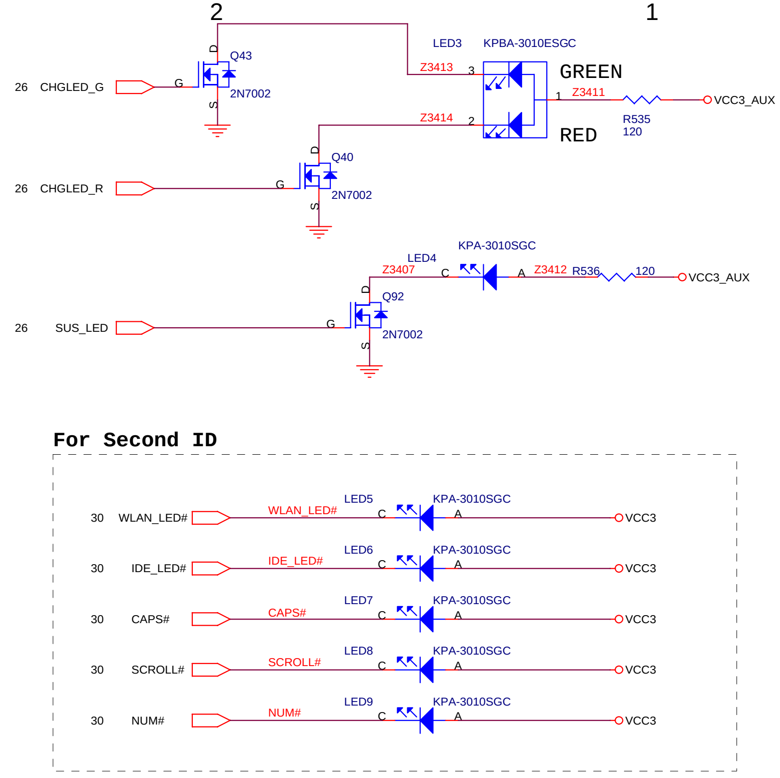
Uniwill International Corp.

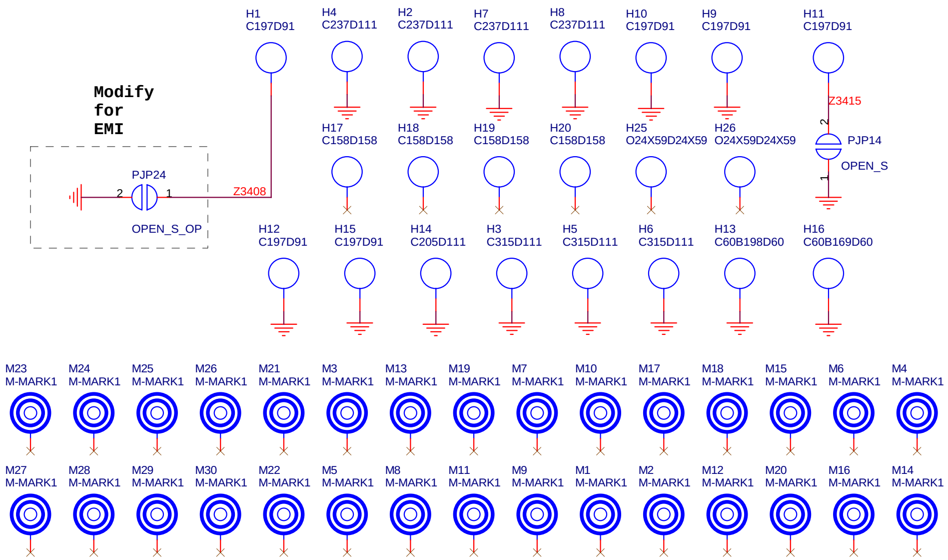
Title	N223II0	
Size	Document Number: 2665 — Battery Conn & Charger	Rev 01
Date: Tuesday, May 25, 2004	Sheet 33 of 39	

Intel PRO/Wireless 2100 LAN

PIN11	LED_WLAN_LINK	Hi(3.3V) Low(0V)	Solid ON 1 flash/3 sec LED OFF	Associated AP Not Associated with an AP Power OFF or RF_Kill active
PIN12	LED_WLAN_ACT	Hi(3.3V) Low(0V)	Rapid Blinking Slow Blinking LED OFF	Passing data traffic to AP Beacon traffic to Ap Power OFF or not activity or RF_Kill active
PIN13	HW_RadioXMIT_OFF#	Hi(3.3V) Low(0V)	Enable Disable	Radio transmitter is ON Radio transmitter turn off







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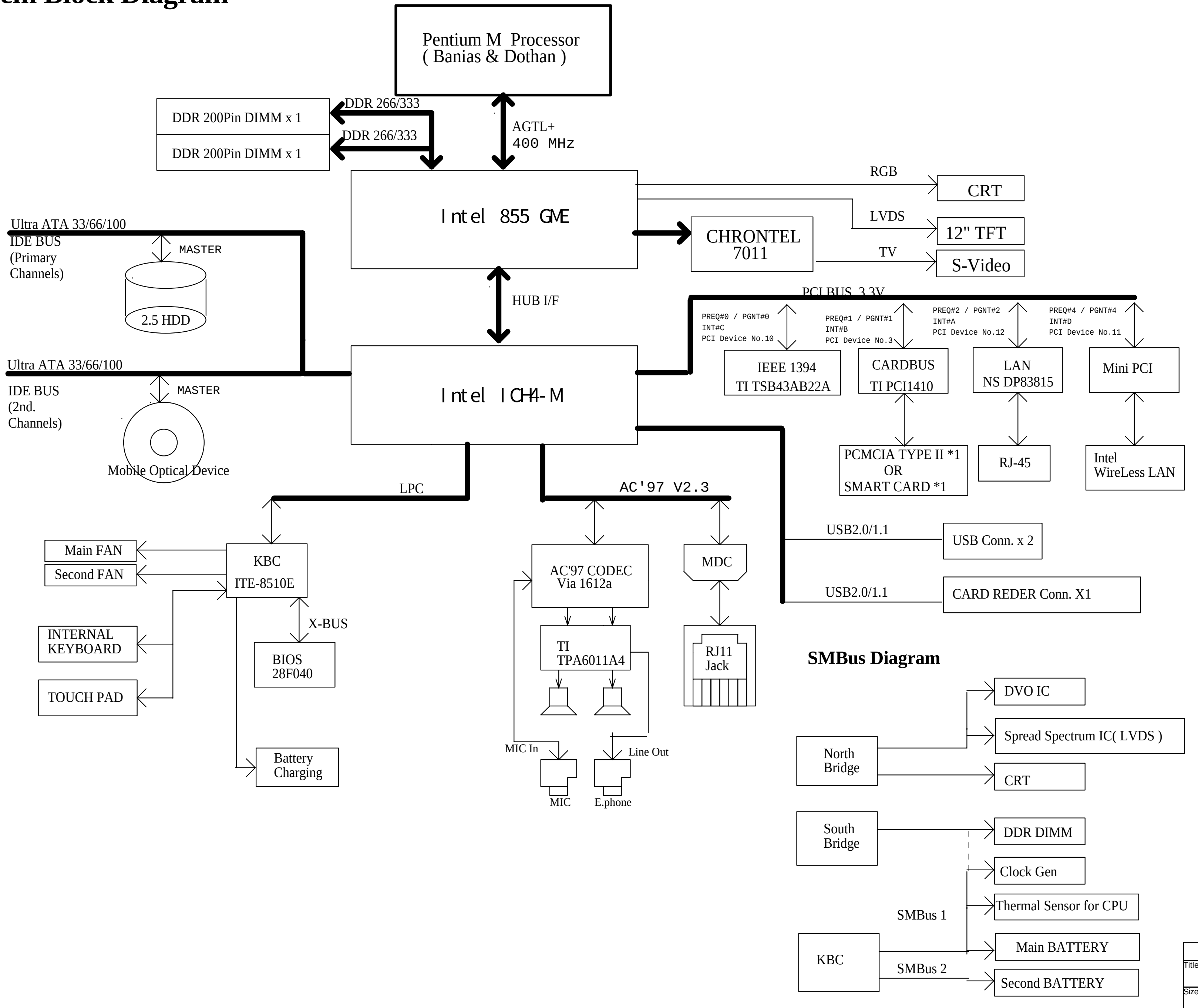
System Block Diagram

INTA# : X
INTB# : PCMCIA
INTC# : 1394
INTD# : LAN

PREQ#0 : 1394
PREQ#1 : PCMCIA
PREQ#2 : LAN
PREQ#3 :
PREQ#4 :



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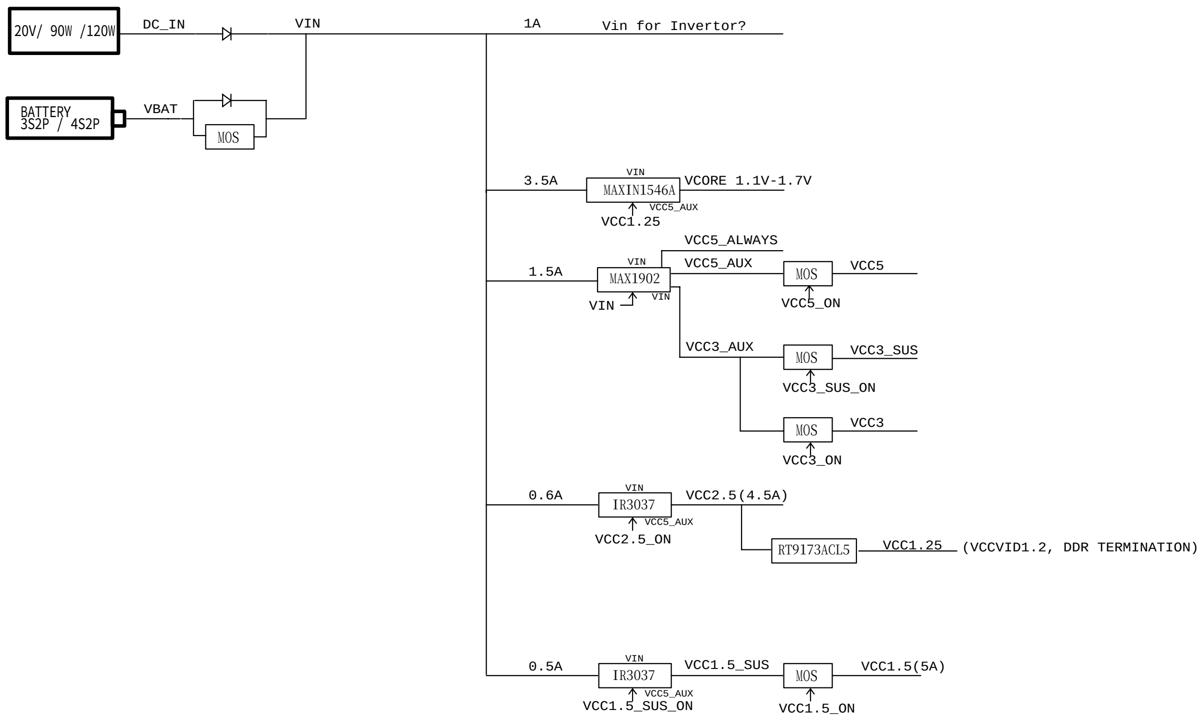
POWER FLOW CHART



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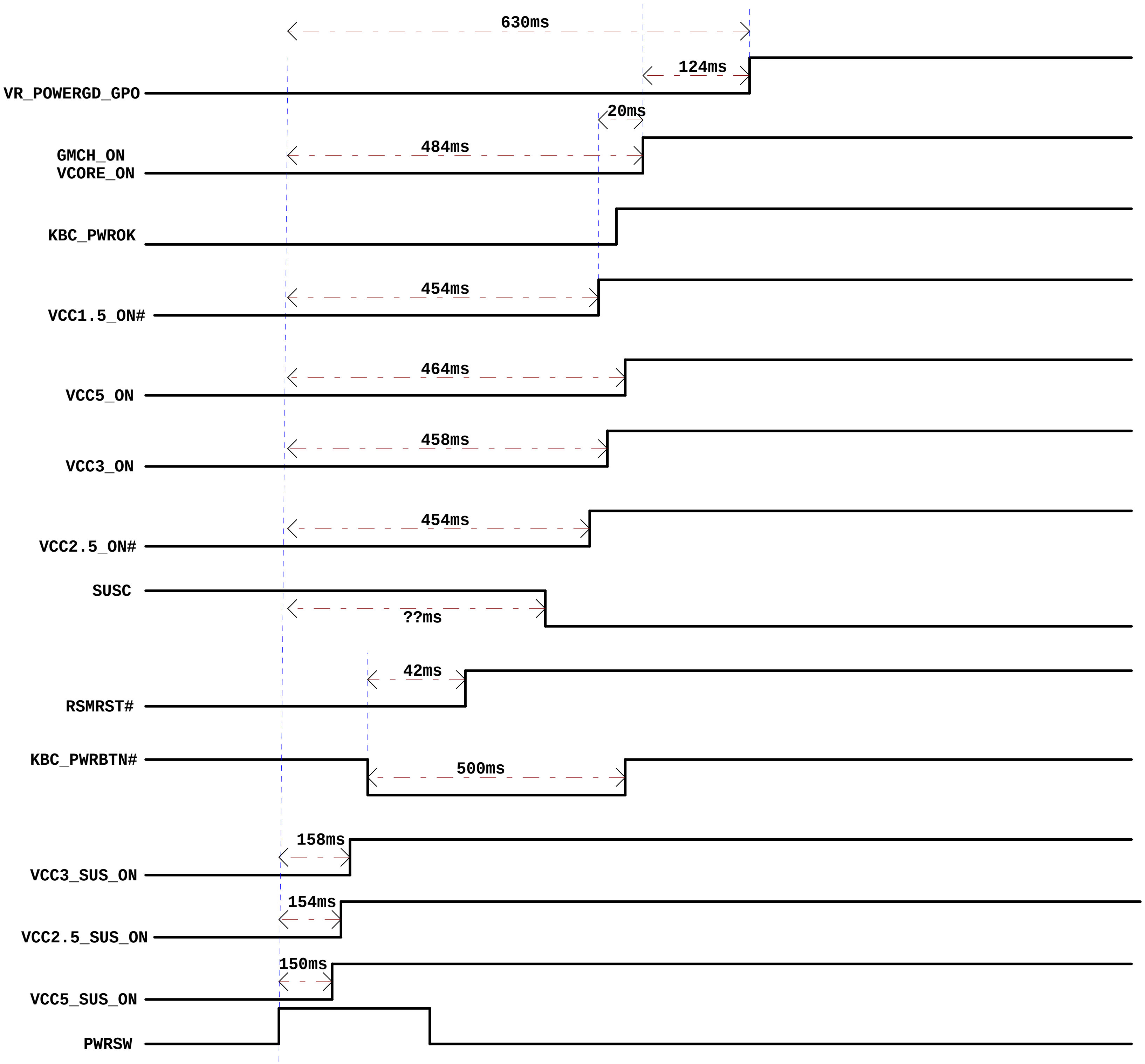
R:C to R:1 Change Note

Modify 1: Page 04 , change SI4362 to IRF4832 to improve power efficiency .
Modify 2: Page 04 , change R561 to 0 ohm to set PWM Frequcecy to 200K.
Modify 3: Page 20 , Del C11 & C14 to fine tune TV signals.
Modify 4: Page 25 , Del B64 for MIC test.
Modify 5: Page 06 , reserve over voltage circuit for over clocking use .
Modify 6: Page 11 , change RTC circuit for fine tune RTC clock .
Modify 7: Page 31 , change R454 to 40 mohm to total power measure setting .
Modify 7: Page 31 , add Q97 to down Banias celeron voltage in low frequency mode .
Modify 8: Page 26 , add R530 to down DDR performerce when DDR over temp.
Modify 9: Page 11 , change C346 value to 12p to fast RTC timing.
Modify 10: Page 17 , change C509 value to 1U.



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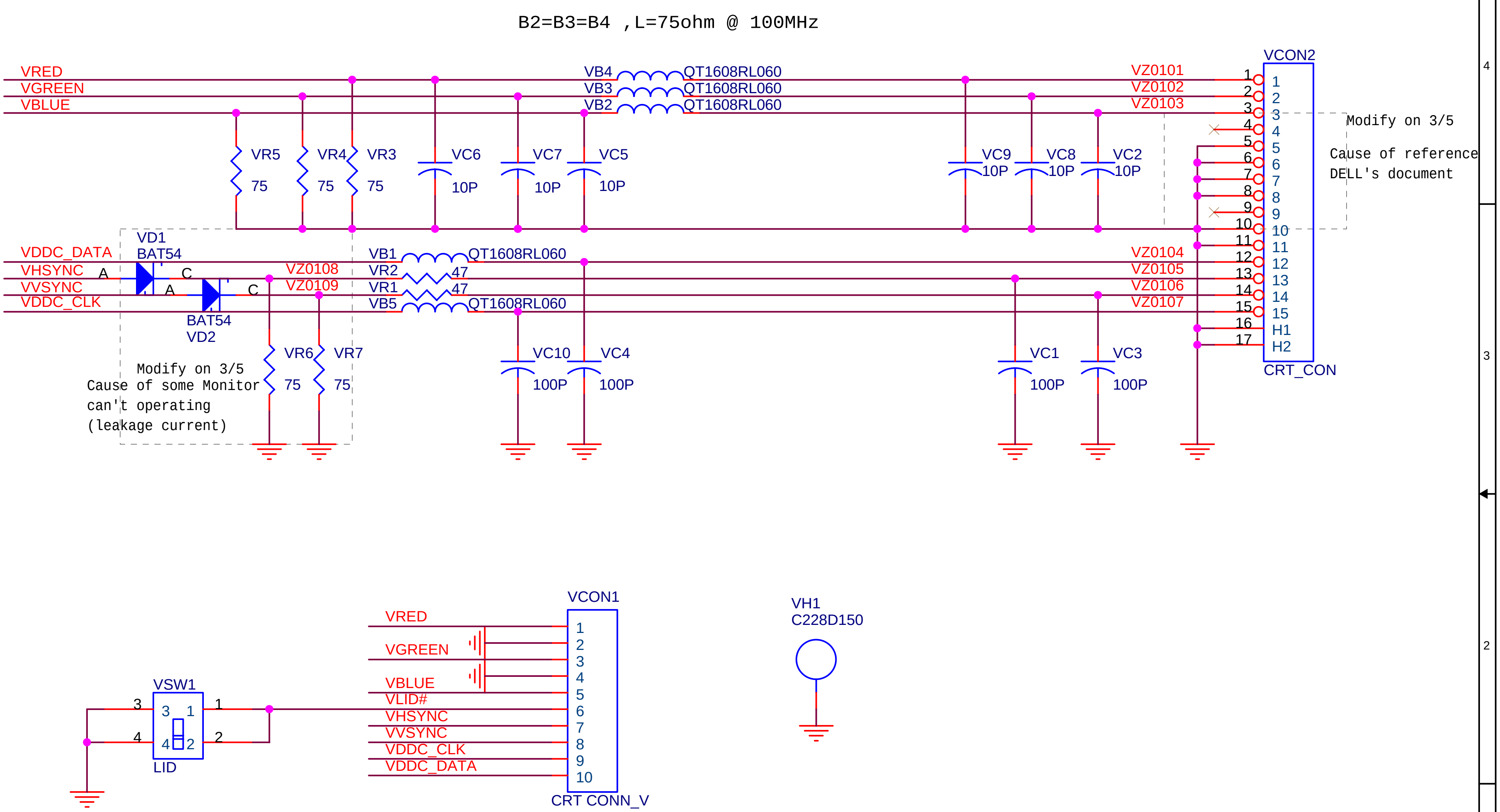

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